The OCP across languages: An argument for Harmonic Layer Theory

Sören E. Tebay, Jochen Trommer, and Eva Zimmermann

January 2025

Contents

1 Introduction

It is a well-established fact that different phonological behaviours can co-exist within the same language. In this paper, we focus on what we call 'interstratal differences', namely patterns where the morpho-syntactic domains of stem, word, and phrase show different phonological patterns within the same language. The introductory examples in (1) all face the same phonological problem, namely two high-toned tone-bearing units that are expected to become adjacent and thus form a marked structure. In Tiriki, this problem is avoided by dissimilating the second of the two underlying high tone (=H-tone) into a low tone (=L-tone) within words (1-a-i). Across words, however, the two adjacent H-tones happily surface (1-a-ii). And whereas dissimilation between two H-tones can be observed within stems in Dagaare (1-b-i), there is no dissimilation but downstep creation for word-sized domains (1-b-ii). Similarly, in Bari a sequence of two H-toned syllables is tolerated within stems (1-c-i) but repaired by dissimilation within words (1-c-ii).

(1) Different repairs for OCP(H) in different domains

a. Tiriki (Paster & Kim, 2011, 79,86)

(i) Dissimilation within words
xu-mú-rhúm-a → xúmúrhùma
INF-3SG.O-send-FV
'to send him/her'

(ii) No repair within phrases
... zí-ꜜsíímbwá lí-ꜜdúúmà → ... zí-ꜜsíímbwá lí-ꜜdúúmà
NC-dog NC-corn
'[he is giving] dogs corn'

b. Dagaare (Anttila & Bodomo, 2022, 445,460)

(i) Dissimilation within stems
búú-ró → búú-rò
measure-IPFV
'measure'

(ii) Downstep within words
... kúlí-lá ... → ... kúlí-ꜜlá ...
... go.home.PFV-FOC ...
'He has gone home'

c. Bari (Yokwe, 1986, 172,173,254)
   (i) No repair within stems
      kàmú-tát → kàmú-tát
      guest-SG
      'guest'
   (ii) Dissimilation within words
      ná-kópò → ná-kòpò
      of-cup
      'of the cup'

The empirical claim in this paper is that interstratal differences are restricted across languages. More concretely, they obey the *ABA restriction that excludes any language where two non-adjacent morpho-syntactic layers (=stem and phrase) show the same phonological behaviour but the intermediate layer (=word) shows a different phonological behaviour. The empirical reality of the *ABA restriction is a strong argument against any theory that derives interstratal differences from co-grammars, i.e. classical Lexical Phonology or Stratal OT.

The main theoretical proposal of this paper is a novel and more restrictive theory of the morpho-phonology interface that inherently predicts the *ABA restriction: Harmonic Layer Theory (=HLT). This model shares with comparable stratal accounts a cyclic optimization of stems, words, and phrases but crucially differs from these alternatives since all optimizations within one language are based on the same grammar. There are hence no different co-grammars or subgrammars within a single language. Interstratal differences fall out in HLT from a mechanism of activity decay across layers. If gradient activity determines phonological behavior and activity monotonically decays across cycles, then phonological behavior will change across cycles. More concretely, the same element can act differently depending on the number of phonological cycles it already passed through: Interstratal differences arise. Such a cyclic model with a single grammar is more restrictive than comparable alternatives. Most notably, patterns violating the *ABA restriction are inherently impossible in HLT.

This paper is structured as follows. In section 2, we present the results of a representative typological study on interstratal differences in OCP contexts across languages. In section 3, we conclude that interstratal differences are indeed systematically restricted and formulate the novel *ABA restriction. After contextualizing this restriction within the research on interstratal differences, we conclude that it is both more and less restrictive than existing alternative restrictions. In section 4, we introduce our theoretical proposal: Harmonic Layer Theory. This novel theory of the morpho-phonology interface is put to the test in section 5 where we first illustrate how the theory derives the interstratal differences we found in our typological sample by presenting three representative in-depth case studies: Shona 5.1, Dagaare 5.2, and Bari 5.3. In subsection 5.4, we return to the *ABA restriction and show that HLT inherently predicts it from its architecture: It is impossible to derive interstratal differences violating the *ABA restriction in HLT. Section 6 concludes and gives an outlook to future research.

2 A typology of interstratal differences in OCP contexts

We investigated the typology of attested interstratal differences in the empirical domain of expected adjacent identical tones (cf. (1-a)), henceforth 'OCP contexts'. Such contexts are an ideal testing ground for such an endeavour since languages vary considerably in whether they allow this marked structure or whether they repair it via one of many possible repair strategies. A list of possible behaviours that could surface in an OCP context is given in (2) within a toy OT-tableau filled with some standard constraint on tones (Yip, 2002) that will resurge again in the theoretical accounts below. Its input contains two adjacent tone-bearing units (=TBU's) that are both associated to their own H-tone. Although our investigation of OCP contexts was a general one, we only found examples of interstratal differences with adjacent H-tones. Although most of the following discussion also applies to other sequences of identical adjacent tones, we will hence focus on H-tones. These two adjacent identical underlying tones can simply be realized as such (2-a) under violation of OCP-H or can be repaired by any of the four repair strategies in (2-b-e). Deletion of one H-tone (2-b) induces a violation of MAX-H and of SPECIFY since a TBU remains without a tone. This SPECIFY violation is avoided if an epenthetic L-tone associated to the TBU instead (2-c). This, however, induces an additional violation of DEP-L. In the following, we will refer to this strategy as 'dissimilation' since it changes an underlyingly H-toned TBU into an L-toned one. For both dissimilation and deletion, there is of course the further variation of whether the first or the second of the two H-tones changes – this question is largely ignored in the following since we did not encounter an example where the target of the same process changed across layers within the same language. In (2-d), both TBU's keep their underlying H-tone but an inserted L-tone remains floating inbetween them: We take this to be the representation of a downstepped H-tone (cf. e.g. Paster & Kim, 2011). Candidate (2-e), finally, also keeps both underlying H-tones but coalesces them into a single one. This repair is trickier than others since it is in fact surface-identical to the OCP-violating structure (2-a). We will hence only assume that fusion applied if there is some independent argument for it, namely if the resulting H-toned TBU's behave as a single unit for other processes. An argument along these lines is made in Shona in section 5.1.

(2) OCP context: Possible behaviours

	H H		OCP-H	Max-H	Dep-L	Unif	Spec	$*{}^{\downarrow}$H
a.	H H	No repair	-1					
b.	H	Deletion		-1			-1	
c.	H L	Dissimilation		-1	-1			
d.	H L H	Downstep			-1			-1
e.	H	Fusion				-1		

For our typological sample of interstratal differences in OCP contexts, we searched for languages that show at least two of the different behaviours in (2) in different morpho-syntactic domains. Such a classification naturally relies on a detailed morphological and phonological analysis to, for example, be sure that the distinction into stem and word layer affixes is not at odds with any independent morphological or phonological facts. In some cases, such an argumentation was already provided in the literature (examples are Shona or Kishambaa), for other languages, we conducted such analyses ourselves (an example is Bari). Many of the languages in our sample are, in fact, Bantu languages where the distinction between words and (macro-)stems is well-established based on a variety of other facts (e.g. the fact that H-tone spreading or reduplication is also sensitive to stem boundaries). For all languages of our sample that are not discussed in section 5, the sources, relevant data, and background facts are given in the appendix 7. Our qualitative sample includes 13 languages so far, all listed with some typological details in (3).

(3) Languages: Typological details (Hammarström et al., 2019)

1.	Jita	jit	Atlantic-Congo	
			East Bantu->Northeast Savanna	Tanzania
2.	Kishambaa	ksb	Atlantic-Congo	
			East Bantu->Northeast Savanna	Tanzania
3.	Chizigula	ziw	Atlantic-Congo	
			East Bantu->Northeast Savanna	Kenya+Tanzania
4.	Tiriki	ida	Atlantic-Congo	
			East Bantu->Northeast Savanna	Kenya
5.	Shona	sna	Atlantic-Congo	
			East Bantu->Southern Bantu	Zimbabwe
6.	Tswana	tsn	Atlantic-Congo	
			East Bantu->Southern Bantu	Botswana
7.	Venda	ven	Atlantic-Congo	
			East Bantu->Southern Bantu	S. Africa+Zimbabwe
8.	Dagaare	dga	Atlantic-Congo	
			Gur->Northern Central Gur	Ghana
9.	Dagbani	dag	Atlantic-Congo	
			Gur->Northern Central Gur	Ghana+Togo
10.	Moore	mos	Atlantic-Congo	
			Gur->Northern Central Gur	Burkina Faso
11.	Chitonga	toi	Atlantic-Congo	
			East Bantu->Botatwe	Zambia
12.	Bari	bfa	Nilotic	
			Eastern Nilotic	South Sudan
13.	Moro	mor	Heibanic	
			West-Central Heibanic	Sudan

The interstratal differences in OCP contexts we found for these languages are summarized in (4). It can be seen that the 13 languages employ 8 different patterns of OCP reactions across layers. Even though the sample is not at all balanced when it comes to areal or genealogical distribution, it is still qualitatively representative in proving a substantial variation of patterns. All the possible behaviours in (2) are represented in our sample.

(4) The typology of OCP repairs

	① Stem	② Word	③ Phrase
Moro	Deletion	No repair	No repair
Shona	Fusion	Deletion	No repair
Chitonga	Deletion	Deletion	No repair
Dagaare	Dissimilation	Downstep	No repair
Tiriki	Dissimilation	Dissimilation	No repair
Moore	Dissimilation	Dissimilation	No repair
Dagbani	Dissimilation	Dissimilation	No repair
Jita	Deletion	Deletion	No repair
Chizigula	Deletion	Deletion	No repair
Kishambaa	Fusion	Downstep	Downstep
Tswana	Fusion	Fusion	Downstep
Venda	Fusion	Fusion	Dissimilation
Bari	No repair	Dissimilation	Dissimilation

3 The *ABA restriction

A *ABA-restriction is well-known from research in morphological typology. Bobaljik (2012) notes a gap in the morphology of comparative and superlative adjective forms, which he terms *ABA. He finds that syncretism between the superlative form and the positive form to the exclusion of the comparative form is not attested. Similar gaps have been found in further areas of grammar, e.g. case syncretism (Caha, 2009), pronoun suppletion (Smith et al., 2019), the ban on improper movement, William's cycle, omnivorous number (Graf, 2020), and person case constraints (Graf, 2019). In this paper, we now formulate a phonological *ABA-restriction which excludes certain interstratal differences. Even though the languages in our typological survey employ a surprising variation in their OCP contexts (4), there is one systematic restriction that holds over all attested patterns. It becomes most apparent if we summarize the patterns in (4) into abstract patterns that only distinguish phonological 'behaviour' as either the absence of any process or the application of a specific process. Only four of such general patterns of phonological behaviour are attested: Three different behaviours at ①, ②, and ③ (5-A), a certain behaviour at ① and different one at ② and ③ (5-B), or an identical behaviour at ① and ② and a different one at ③ (5-C).

(5) General patterns of interstratal differences in our sample

	Pattern A	Pattern B	Pattern C
①	Behaviour 1	Behaviour 1	Behaviour 1
②	Behaviour 2	Behaviour 2	Behaviour 1
③	Behaviour 3	Behaviour 2	Behaviour 2
	3 lgs	7 lgs	3 lgs

A general pattern that is systematically unattested is hence an ABA-pattern where one phonological behaviour is attested at the stem and the phrase layer but another behaviour at the intermediate word layer. We formulate this as the *ABA restriction (6). All the patterns we found in our representative typology of OCP contexts (61) are compatible with this restriction.

(6) The *ABA restriction
In a given language L: It is impossible that the same phonological context shows phonological behaviour P1 in layer ① and ③ but a different phonological behaviour P2 in the intermediate layer ②.

Apart from our representative typological study, the *ABA restriction receives further support from the existing literature and the formulation of related restrictions. Since the advent of Lexical Phonology (Kiparsky, 1982, 1985), researchers wondered whether and how interstratal differences can and should be restricted. The predictive power of Lexical Phonology that allows to combine completely different co-grammars within the same language was hence deemed to be to powerful and in need of some restriction. The first formulation of such a restriction we are aware of is the Continuous Stratum Hypothesis (7)[1] which excludes phonological rules which apply only in non-contiguous strata. It is entailed in the more restrictive Strong Domain Hypothesis (8) which states that rules can be turned 'off' but they cannot be turned 'on' in non-initial strata.

(7) Continuous Stratum Hypothesis (Mohanan, 1982, 1986)
The domain of a rule is specified as a set of continuous strata

[1] It also surfaced under the terms 'Stratum Domain Hypothesis' or 'Continuity of Strata Hypothesis' in the work of Mohanan.

(8) Strong Domain Hypothesis (Kiparsky, 1984, 1985)
A phonological pattern enforced in any given level must also be enforced at all earlier levels.

After the proposal of the Strong Domain Hypothesis, various empirical arguments were put forward that it is too restrictive. One of the most convincing arguments are probably the discussion in Inkelas & Orgun (1995) on Turkish velar deletion or the one in Hyman (1993) on Dagbani H-Spreading. We also found a counterexample to the Strong Domain Hypothesis in our typology, namely Bari which is discussed in more detail in subsection 5.3.

To our knowledge, there is no empirical counter-argument against the Stratum Domain Hypothesis. Beside the question of empirical reality, there is of course the question of how to implement any potential restriction on interstratal differences into a theoretical model. Both (7) and (8) were restrictions super-imposed on Lexical Phonology, they did not fall out from anything inherent in the proposal. Furthermore, as is, for example, explicitly discussed in Inkelas & Zoll (2007), such restrictions are not easily formalizable in Stratal OT (Bermúdez-Otero, 2012; Kiparsky, 2015; Trommer, t.a., 2025), the optimality-theoretic descendant of Lexical Phonology. In terms of interstatal differences, Stratal OT is hence maximally unrestricted and in principle allows any combination of grammars for stems, words, and phrases within the same language. We argue that this is too much predictive power and that interstratal differences across languages are indeed systematically restricted by the *ABA restriction.

At first glance, the *ABA restriction seems identical to the Continuous Stratum Hypothesis since it bans discontinuos patterns. However, there is the subtle but crucial difference that the *ABA restriction refers to phonological behaviour in general and not to specific phonological processes. It hence bans the abstract configurations in (9). Pattern A where the outer layers don't show any phonological repair in a given context but the intermediate layer shows a certain process is a pattern that is indeed possible under the Continuous Stratum Hypothesis but not the *ABA restriction.

(9) Excluded by the ABA-restriction

Layer	Identical context C: *Pattern A	*Pattern B	*Pattern C
Stem ①	no repair	process P1	process P1
Word ②	process P1	no repair	process P2
Phrase ③	no repair	process P1	process P1

The *ABA restriction is hence more restrictive than the Continuous Stratum Hypothesis. On the other hand, it is less restrictive than the Strong Domain Hypothesis since it indeed allows patterns like (10); pattern E being exemplified with Bari (cf. section 5.3).

(10) Possible under *ABA but excluded from Strong Domain Hypothesis

	Identical context C: Pattern D	Pattern E	Pattern F
①	no repair	no repair	no repair
②	no repair	process P1	process P1
③	process P1	process P1	process P2

Most importantly, however, we argue that the *ABA restriction is not a restriction that needs to be super-imposed unto a theory of interstratal differences. In fact, the empirical reality of the *ABA restriction is the main argument for our novel proposal of Harmonic Layer Theory. In this model, interstratal differences arise from cyclic optimization with a single phonological grammar for each language and the novel representational assumption of activity decay across layers. As is laid out in detail in the section 5.4, the *ABA restriction is an inherent property of HLT.

4 Theoretical Background: Harmonic Layer Theory

HLT is a novel model of the phonology-morphology interface that relies on three main assumptions: A. The phonological component of grammar is organized in three morphological cycles, B. output candidates are evaluated against a single phonological grammar, and C. all phonological objects have a certain gradient activity (Trommer, 2019; Zimmermann & Trommer, 2022;

Zimmermann, 2024, 2025; Tebay, 2023, 2025).

Assumption A. is shared between HLT and classical Lexical Phonology (Kiparsky, 1982, 1985) or stratal OT (Bermúdez-Otero, 2012; Kiparsky, 2015; Trommer, t.a., 2025). It assumes a cyclic interaction of morphology and phonology where the concatenation of certain morphemes is interleaved with phonological evaluations. In a classic model with three layers that we also adopt here, roots and stem-layer affixes form first stems which are then subject to a first phonological evaluation. The output of this first cycle of phonology is then fed back into the formation of words which in turn is fed back into the phonology before phrases are formed which in turn undergo a final cycle of phonological evaluation.

Assumption B crucially sets stratal models and HLT apart: It is the same phonological grammar that optimizes cyclically within a language. This is similar to other cyclic single grammar approaches, such as Harmonic Serialism (McCarthy, 2010, 2016). In contrast, Lexical Phonology and Stratal OT assume that potentially different grammars can evaluate at the stem, word, and phrase layer which in turn results in interstratal differences.

In HLT, the source for interstratal differences lies in assumption C, namely the assumption of gradient activity for phonological elements. More concretely, all linguistic symbols have a gradient activity in underlying representations and a — potentially different —- surface activity (e.g. Smolensky & Goldrick, 2016; Rosen, 2016, 2019a,b). Since different activities can result in gradient constraint violations of both faithfulness and markedness constraints (Zimmermann, 2018, 2019, 2021, to appear; Faust & Smolensky, 2017; Jang, 2019; Walker, 2019), seemingly identical phonological elements can show distinct phonological behaviour. The assumption of underlying activity differences is hence an explanation for lexical exceptionality as has been worked out in the existing research on gradient activity (e.g. Zimmermann, 2019). HLT explores a different prediction that any model based on gradient activity makes: That activity can predictable be adjusted as a phonological repair, briefly illustrated in tableau (12). A context-free markedness constraint *H (11-a) penalizes all H-tones. Since it its violations are evaluated gradiently and the constraint is hence sensitive to the activity of a H-tone, a weaker H-tone will violate it to less severely than a more active tone. Adjusting activity is taken to be a possible phonological repair which is penalized by specific faithfulness constraints like IDA for tones (11-b). In addition to general IDA constraints, we assume that there are threshold constraints penalizing an activity adjustment that goes beyond a certain set amount. In our toy example (12), this amount is set to 0.2 by the constraint (11-c). The contrast between the constraints (11-d) and (11-e) illustrates a further assumption we make about the constraint inventory of HLT: Every markedness and faithfulness constraint exists in a gradient and a categorical version. The standard MAX constraint against deletion hence exists in a version that is violated depending on the input activity of the deleted element (11-d) and in a version that is insensitive to any activities (11-e). The latter constraints are always given in double straight bars in the following. Section 5 will emphasize how the contrast between categorical and gradient constraints predicts interstratal differences. Finally, as all models based on gradient representation, the phonological optimization in the following is based on Harmonic Grammar and thus weighted constraints (Legendre et al., 1991, 2006; Pater, 2009; Potts et al., 2010). In such models, a harmony score ($\mathcal{H}$) is calculated for each candidate that adds all constraint violations times their respective constraint weights and the candidate with the harmony score closest to zero is optimal.

(11)
a. *H
   Assign -x violation for every H tone $H{\bullet}_x$.
b. IDENTACT(H) (=IDA)
   For every H-tone H_y corresponding to input H_x, assign -(|x-y|) violation.
c. IDENTACT(H)>.2 (=IDA≥.2)
   For every H-tone H_y corresponding to input H_x, assign -(|x-y|) violation if (|x-y|)>0.2.
d. MAX-H
   Assign -x violation for every input H-tone H_x without a corresponding output tone.
e. ‖MAX-H‖
   Assign -1 violation for every input H-tone without a corresponding output tone.

Tableau (12) shows the effect of these constraints for an input that contains a fully active H-tone /H_1/ and a lexically weaker /$H_{0.8}$/ tone. As can be seen, activity is expressed by subscripted numbers. Faithful realization of these tones in candidate (12-a) induces -1.8 violations of the markedness constraint against H-tones: The combined activity of all H-tones in the output. Reducing the activity of both tones to the minimal activity of 0.1 (12-b) reduces the violations of *H to -0.2 but induces violations of both IDA and IDA≥.2. The latter is violated since the activity reduction for both tones exceeds the threshold of 0.2. This violation is avoided in candidate (12-c) that only reduces the activity of both tones by 0.2 and consequently only violates the general IDA. This candidate emerges as the winner since it manages to avoid as many violations of *H without inducing a violation of high-weighted IDA≥.2. Candidates (12-d) and (12-e) show that the complete deletion of a H-tone is of course the ideal solution to avoid any violations of *H but that this strategy is excluded in our toy grammar. The contrast between the violations of (12-d) and (12-e) illustrates the difference between categorical ‖MAX-H‖ and gradient MAX-H: The latter will always favour deletion of a less active input element over deletion of a stronger input element.

(12) Activity loss for tones

H_1 $H_{0.8}$	IDA≥.2 11	MAX-H 11	‖MAX-H‖ 11	*H 10	IDA 1	$\mathcal{H}$
a. H_1 $H_{0.8}$				-1.8		-18.0
b. $H_{0.1}$ $H_{0.1}$	-1.6			-0.2	-1.6	-21.2
☞ c. $H_{0.8}$ $H_{0.6}$				-1.4	-0.4	-14.4
d. $H_{0.6}$		-1	-1	-0.6	-0.2	-28.2
e. $H_{0.8}$		-1	-0.8	-0.8	-0.2	-28.0

In a system based on gradient representations, activity adjustment is hence an expected phonological process. And in a cyclic model where a grammar like (12) optimizes increasingly larger morphological domains, elements are predicted to gradually adjust their activity across layers, as shown in (13) for a cyclic optimization based on (12). A fully active H-tone that enters the stem optimization will hence be optimized three times before it reaches the final phrase output and will loose 0.6 activity in total: A small amount of decay at each optimization cycle.

(13) Activity decay across layers, predicted from grammar (13)

Optimization							
① Stem	$/H_1/$	→	$H_{0.8}$				
② Word			$/H_{0.8}/$	→	$H_{0.6}$		
③ Phrase					$/H_{0.8}/$	→	$H_{0.4}$

And given the basic tenet of any model based on gradient activity that activity differences can result in different phonological behaviour, predictable activity decay in a cyclic model inherently predicts interstratal differences. Crucially, interstratal differences fall out from cyclic optimization with the same grammar; the only reason for different phonological behaviour are activity differences in representations. The three case studies in the next section 5 are all based on the exact assumptions of (13), namely that all tones decay predictably by 0.2 at each optimization cycle. That the languages differ in their concrete interstratal differences falls out from the constraints sensitive to gradient activity that have an influence in a given grammar.

Before proceeding, however, we want to emphasize that any decay for a phonological element within HLT is language-specific since it is a consequence from a specific grammar. There is hence no built-in mechanism that derives activity adjustments; it is simply a phonological process. In a language without any interstratal difference (for certain phonological elements), there is hence no need to assume any activity decay across cyclic evaluations.

5 Interstratal differences in HLT

In this section, we now show in detail how HLT predicts interstratal differences from the basic mechanism of activity decay alone. First, we present three representative case studies from our typology in subsections 5.1 to 5.3. The empirical patterns in Shona, Dagaare and Bari are quite distinct (14) but their HLT accounts are all based on the same basic mechanism, namely predictable activity decay for tones. Recall from above that we assume that tones decay by 0.2 at each optimization in all three of these languages (cf. tableau (12)), sketched again in (14). This activity decay will then result in less violations of certain gradient constraints at the next layer. The languages will only differ in which gradient faithfulness or markedness constraints are relevant to predict the respective patterns.

(14) Interstratal differences in OCP contexts: Three case studies

	Shona	Dagaare	Bari
①	Fusion	Dissimilation	No repair
②	Deletion	Downstep	Dissimilation
③	No repair	No repair	Dissimilation

(15) Tone decay in Shona, Dagaare, and Bari

	Input	Output
①	/H_1/	$H_{0.8}$
②	/$H_{0.8}$/	$H_{0.6}$
③	/$H_{0.6}$/	$H_{0.4}$

5.1 Fusion, deletion, or no repair in Shona

The interstratal differences in OCP contexts in Shona have been discussed in a seminal paper by Myers (1997). In this paper, an argument for an OT approach with a violable OCP constraint is made based on the language-internal and -external variation in OCP contexts. The Shona case study is particularly illuminating since the language shows three different repairs at the three layers stem, word, and phrase and those crucially interact with each other. All data and generalizations in the following are taken from Myers (1986) and Myers (1997) where the Zezuru dialect of Shona, spoken in Zimbabwe, is described.

5.1.1 Data

The TBU in Shona is taken to be the syllable since there are no effects of vowel length on possible tone combinations within a syllable. There are only H-toned (=v́) and L-toned (=v) TBU's and no contour tones. As is typical for Bantu languages, Shona is taken to be best analysed as an underlying contrast between TBU's associated to H-tones and tone-less TBU's, there are hence no underlying L-tones.

Shona shows the typical Bantu verb structure, where the verb root, all suffixes, and certain prefixes form a (macro-)stem under exclusion of other pre-poned elements. In Shona, all object prefixes and some tense and subject prefixes are part of the stem, other subject and tense-markers are outside of the stem domain and are assumed to form a stem on their own (see also Jones, 2014), cf. (16). I follow the convention in Myers (1997) and notate stems with square brackets.

(16) Morphological structure of Shona verbs

Word				
	[Macrostem]	[Macrostem]		
	Clitics	prefixes–	root	–suffixes
	(copula, Assoc., remaining inflection)	(object, some subject, some Tns)		
e.g.	ku-	mú-	téng	-és-ér-a
	INF	OBJ	buy	CAUS-applied-FV

Another relevant background fact about the language is the pervasiveness of H-tone spreading. As can be seen in (17), underlying H-tones spread rightward to following otherwise tone-less TBU's – I follow Myers (1997) in using underlining to mark vowels that are best analysed as underlyingly H-toned. In fact, H-spreading also shows interstratal differences: H-tones spread to the next two following TBU's at the stem layer (17-a) but only to the following TBU at the word and phrase layer (17-b).[2] In most of the examples below, these underlying H-tones spread and cause more TBU's to be H-toned.

(17) H-spreading in Shona

a. [ku]-[téng-és-á]
INF-buy-CAUS-FV
'to sell' (Myers, 1997, 856)

b. [í]-[sádza]
COP-porridge
'(it) is porridge' (Myers, 1997, 860)

With this background, we can now turn to the OCP contexts in Shona. We start with the phrase layer since the behaviour is most transparent here: Adjacent H-toned TBU across word boundaries are happily tolerated at the phrase layer as can be seen in (18) where a H-final noun precedes a H-initial adjective.

[2] These facts can easily be implemented from H-tone decay in HLT as well if the triggering constraint demanding H-Spread is not violated enough anymore at the word and phrase layer to warrant longer spreading. To streamline our argument, we decided to focus on interstratal differences with regard to the OCP.

(18) OCP tolerated across words ③
badzá gúrú
hoe big
'big hoe' (Myers, 1997, 874, FN.21)

Within words and between stems, however, the second of two otherwise adjacent H-tones is deleted as is shown in (19). This deletion process is famous in Bantu as 'Meussen's Rule'. In (19-a), a H-toned future prefix is added to a stem starting with a H-toned TBU and this second TBU becomes L-toned on the surface. That the root 'buy' is indeed H-toned was shown above in (17-a) where its H-tone spread to the following suffixes. Example (19-b) adds an example where a H-toned copula is added to a H-initial noun.

(19) Deletion between stems ②
a. [ndi-chá]-[teng-es-a]
1.sg-fut-buy-caus-FV
'I will sell' (Myers, 1997, 856)
b. [í]-[banga]
cop-knife
'(it) is a knife' (Myers, 1997, 856)

Within stems, the picture is once again different: Adjacent H-toned elements seemingly keep their H-tones as can be seen in (20) where H-toned stem layer prefixes are added to H-toned roots and both these morphemes surface with a H-tone.

(20) Fusion at ① between Obj+root
a. [tí-téng-és-é]
1pl/subj-buy-caus-FV
'we should sell' (Myers, 1997, 870)
b. [ku]-[mú-téng-és-ér-a]
inf-obj-buy-caus-applied-FV
'to sell him/her' (Myers, 1997, 869)

The behaviour of stems like (20) in other contexts gives striking evidence for the fact that the distinct underlying H-tones indeed did not surface as such but are subject to fusion instead. The forms in (20) contain hence only a single H-tone, which is associated to multiple TBU's. One piece of evidence for this comes from the interaction with deletion at the word layer.[3] If a stem like (20-a) is preceded by a H-toned word layer affix, all four otherwise H-toned stem-TBU's surface as L-toned as can be seen in (21-a).

(21) Deletion within words ②, fed by earlier fusion ①
[há]-[ti-tenges-e]
HORT-1PL/SUBJ-buy-CAUS-FV
'let us sell' (Myers, 1997, 870)

This fell-swoop deletion receives an immediate explanation if the second stem in (21) only contains a single H-tone that is associated to multiple TBU's: Deletion of this one H-tone causes all of these TBU's to becomes L-toned. This argument relies on the fact that the OCP restriction in Shona excludes adjacent H-toned TBU's, not adjacent H-tones. That this is indeed the case can be seen in data as (22) where two H-toned stems are combined but both H-tones happily surface since they are separated by a tone-less TBU. Returning to (21): If the H-tones in the stem _tí-téng-és-é_ would have remained distinct H-tones, we would wrongly predict *_há-ti-téng-és-é_ at the word layer where only deletion of the second but not the third H-tone already resolves the OCP-problem. Only fusion of adjacent H-tones at the stem layer makes hence the correct prediction. The second example in (19) also illustrates the interaction of deletion and fusion.

(22) No deletion for non-adjacent H's
[í]-[badzá]
COP-hoe
'(it) is a hoe' (Myers, 1997, 860)

[3] The second piece of evidence comes from the interaction with a process of 'tone-slip' or 'tone-retraction' that de-associates a multiply linked first H-tone if it is followed by another H-tone. This process in fact applies at all layers in Shona: Only if retraction cannot solve the OCP-problem do we see the interstratal difference between fusion, deletion, and toleration of the OCP. To streamline the argument and focus on the differences between layers in Shona, we decided to omit this additional empirical fact from the discussion. The theoretical account in section 5.1 can more than easily be extended to incorporate these facts: De-association of an initial H would simply be penalized by a constraint lower weighted than any of the other relevant faithfulness constraints.

5.1.2 HLT account

The empirical pattern of three different reactions to underlying adjacent H-tones in Shona is once again summarized in (23).

(23) OCP contexts in Shona

①	Fusion
②	Deletion
③	No repair

The HLT account for this pattern is based on the assumption that tones decay predictably in Shona at every optimization, derived formally in (12). As was emphasized in section 4, this tone decay is a language-specific consequence that results if a markedness constraint against tonal activity out-weighs the counter-acting faithfulness constraint demanding that all elements retain their activity. That the three different patterns of tone fusion, tone deletion, and toleration of the OCP can co-exist in Shona follows if tone decay interacts with both gradient and categorical markedness and faithfulness constraints. For Shona, the constraint penalizing adjacent H-toned TBU's (24-a) is gradient and hence sensitive to the activity of the tones forming this marked configuration. That Shona only penalizes adjacent H-toned TBU's and not H-tones that are adjacent on the tonal tier became apparent in the discussion of the data in (22). Since tones decay in Shona, an OCP-violation is hence easier and easier to tolerate, the more the tones were already optimized. This is what predicts that OCP-violations are tolerated at the phrase layer. Tone fusion that applies at the stem layer is penalized by a categorical constraint ‖Unif‖ (24-b) and tone deletion that applies at the word layer is penalized by a gradient Max-H (24-c). Since the latter constraint is gradient, its effect will again be less and less fatal the morpho-syntactically more complex the structures get.

(24) a. OCP-•
Assign -$\frac{x+y}{2}$ violation for every pair of H-tones H_x and H_y that are associated to adjacent TBU's.

b. ‖UNIF‖
Assign -1 violation for every pair of input tones corresponding to the same output tone.

c. MAX-H
Assign -x violation for every input H-tone H_x without a corresponding output tone.

There are a variety of additional constraints that exclude other imaginable reactions to an OCP context that simply never surface in Shona. To mention just a few: There are no L-tones in Shona and hence dissimilation as the insertion of an L-tone or downstep as the insertion of a floating L-tone are excluded. In addition, some directional constraint needs to ensure that it is always the second and never the first of two otherwise adjacent H-tones that is deleted at the word layer. Since we are focussing on the interstratal differences within and across languages, we will ignore those alternative strategies as being excluded by higher-weighted constraints.

In tableau (25), we see the optimization for two adjacent input tones at the stem layer that can, for example, arise if a H-toned object prefix is added to a H-initial root (20). Both tones have the full activity of 1.0 in the input but will decay to a 0.8 activity in the output. Candidate (25-a) realizes both adjacent H-tones and thus induces -0.8 violations of OCP-H– the mean activity of the two tones that form the marked configuration. Deletion of on tone (25-b) avoids this violation of OCP-H but induces a MAX-H violation. Since gradient faithfulness constraints are sensitive to the input activity, this constraint is violated by -1 given that a fully active $H_{1.0}$ remains without output correspondent. In candidate (25-c) both input H-tones are fused into a single output tone, determinable from the superscripted indices that notate input-output correspondences. This strategy again avoids any violation of OCP-H but induces a violation of ‖UNIF‖. Since this faithfulness constraint is categorical, it will always be violated by -1, irrespective of the underlying input activity of the fused tones. The constraint weightings assumed for Shona predict that this fusion candidate (25-c) becomes optimal at the stem layer.

(25) ① Stem: Tone Fusion

$^{a}\mathrm{H}_1$ $^{b}\mathrm{H}_1$ σ σ	OCP-• 23	MAX-H 16	‖UNIF‖ 14	$\mathcal{H}$
a. $^{a}\mathrm{H}_{.8}$ $^{b}\mathrm{H}_{.8}$ σ σ	-0.8			-18.4
b. $^{a}\mathrm{H}_{.8}$ σ σ		-1.0		-16.0
☞ c. $^{a,b}\mathrm{H}_{.8}$ σ σ			-1.0	-14.0

The next tableau derives the very same context of two adjacent input H-tones but now it considers word-contexts like (19) where the two adjacent H-tones belong, for example, to a tense prefix and a verb root. The crucial difference to (25) is hence that the input tones are already weakened since they already underwent one course of evaluation.[4] The two gradient constraints OCP-H and MAX-H are hence violated to a lesser degree than they were in the comparable candidates in (25). This seemingly subtle difference has major consequences for the Harmonic Grammar evaluation. In candidate (26-a), OCP-H is only violated by -0.6 now since both tones are weaker than they were at the stem layer (25-a). And deletion in candidate (26-b) only induces -0.8 violations of MAX-H. Both candidates hence have a relatively better harmony score than the corresponding candidates in (25). Fusion in candidate (26-c), on the other hand, retains its harmony score since the relevant constraint is categorical. And this harmony score is now worse than the reduced one of candidate (26-b) – deletion emerges as optimal.

[4]That both tones were already optimized once follows the standard assumption about Bantu morpho-phonology that verbs are formed from two macro-stems and that the prefix clusters are stems on their own (cf. (16)).

(26) ② Word: Tone Deletion

$^{a}\text{H}_{.8}\ \ ^{b}\text{H}_{.8}$ $\vert\ \ \ \ \ \vert$ $\sigma\ \ \ \ \sigma$	OCP-• 23	MAX-H 16	‖UNIF‖ 14	$\mathcal{H}$
a. $^{a}\text{H}_{.6}\ \ ^{b}\text{H}_{.6}$ $\vert\ \ \ \ \ \vert$ $\sigma\ \ \ \ \sigma$	-0.6			-13.8
☞ b. $^{a}\text{H}_{.6}$ $\vert$ $\sigma\ \ \ \ \sigma$		-0.8		-12.8
c. $^{a,b}\text{H}_{.6}$ $\wedge$ $\sigma\ \ \ \ \sigma$			-1.0	-14

At the phrase layer (27), the two gradient constraint violations of OCP-H and MAX-H decrease once more: Candidate (27-a) violates the former only by -0.4 and (27-b) the latter only by -0.6. Interestingly, this additional decay once again favors another candidate as optimal, namely (27-a) that tolerates the OCP. That a -0.6 violation of OCP-H in (26-a) is still worse than a -0.8 violation of MAX-H in (26-b) but a -0.4 violation of OCP-H in (27-a) is better than a -0.6 violation of MAX-H in (27-b) follows from the different weights of the two constraints. OCP-H has a higher weight than MAX-H and a 0.2 difference hence has a more serious effect for the former than it has for the latter. This effect of crossing thresholds is illustrated in more detail below in (29).

(27) ③ Phrase: OCP violation tolerated

$^{a}\text{H}_{.6}\ \ ^{b}\text{H}_{.6}$ $\vert\ \ \ \ \ \vert$ $\sigma\ \ \ \ \sigma$	OCP-• 23	MAX-H 16	‖UNIF‖ 14	$\mathcal{H}$
☞ a. $^{a}\text{H}_{.4}\ \ ^{b}\text{H}_{.4}$ $\vert\ \ \ \ \ \vert$ $\sigma\ \ \ \ \sigma$	-0.4			-9.2
b. $^{a}\text{H}_{.4}$ $\vert$ $\sigma\ \ \ \ \sigma$		-0.6		-9.6
c. $^{a,b}\text{H}_{.4}$ $\wedge$ $\sigma\ \ \ \ \sigma$			-1.0	-14.0

The interstratal differences between three different reactions to the same phonological context can hence fall out from the simple mechanism of predictable activity decay as soon as gradient constraint violations are taken into account. Before proceeding, we want to emphasize that the specific weights given in the preceding tableaux are only exemplifying weights that ease the readability of the tableaux. As in standard Harmonic Grammar, the defining assumptions about the grammar of a language are not specific weights but relative weighting differences between constraints. There are in fact numerous imaginable constraint weights that would correctly predict the Shona pattern. To highlight this point, we summarize the crucial weighting relations for Shona in (28) that determine the thresholds between fusion and deletion (28-a) and the one between deletion and the toleration of the OCP (28-b). These weighting relations also make it clear why HLT is essentially based on Harmonic Grammar: An account with ranked instead of weighted constraints is in principle compatible with the assumption of gradient constraint violations but would be unable to predict the shift of constraint effects we see in (28).

(28) Crucial weighting relations: Shona

a. Fusion at ① but deletion at ②

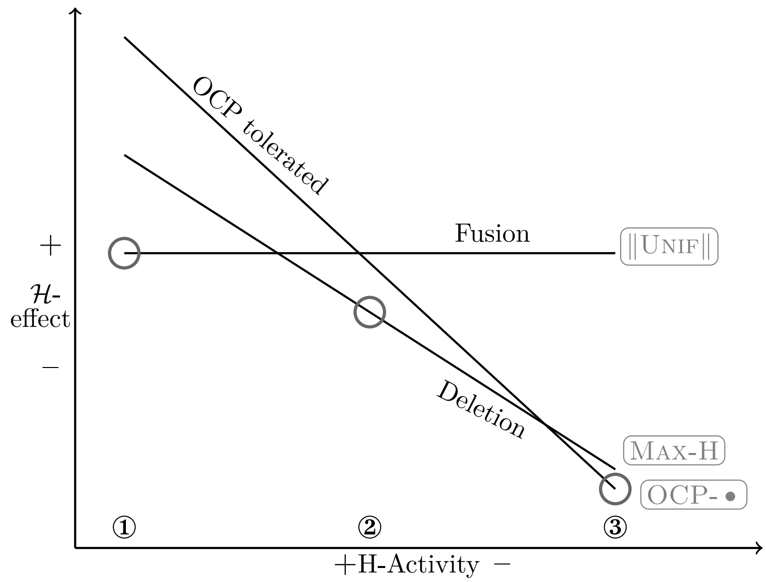
1.0×MAX-H	>	1.0×‖UNIF‖			(25) c>b
		1.0×‖UNIF‖	>	0.8×MAX-H	(26) b>c

b. Deletion at ② but toleration of the OCP at ③

0.6×OCP-H	>	0.8×MAX-H			(26) b>a
		0.6×MAX-H	>	0.4×OCP-H	(27) a>b

Another illustrative way to think about these weighting relations which draws attention to the crossing of harmony score thresholds that fundamentally underlies any HLT account is given in (29). In this graph, the x-axis denotes the tonal activity. The y-axis denotes the harmony score effects that certain candidates induce by their constraint violations. In this graphic representation the fusion candidate corresponds to a straight line: It has the same harmony score effect across layers since the relevant constraint is categorical. The deletion and toleration of OCP candidates, however, correspond to falling lines since their constraint violations decrease. Crucially, the line denoting the toleration candidate shows a steeper fall than the one denoting the deletion candidate because the former is a markedness constraint that refers to output activation, whereas the latter is a faithfulness that makes reference to input activation. The circles denote the least fatal harmony score at each of the three relevant cut-off points for stem, word, and phrasal optimizations. In Shona, three different behaviours emerge since there are three relevant line-crossings.

(29) HLT thresholds in Shona



5.2 Dissimilation, downstep, or no repair in Dagaare

Dagaare is a Gur language spoken in Ghana. Most of the following data is taken from Anttila & Bodomo (2022) (=‘AB’) but we also consulted Angsongna (2023) (‘Ang’). The two works describe slightly different dialects which, however, do not differ in the relevant data cited here. As in Shona, three different behaviours can be observed in OCP contexts across layers. Interestingly, however, the concrete repair operations are different in Dagaare. This subsection hence shows how the same general pattern of two crossing harmony score thresholds (29) can be instantiated by different constraints resulting in different processes.

5.2.1 Data

Dagaare contrasts three lexical tone specifications: H-tones (=v́), L-tones (=v̀), and downstepped H-tones (=ꜜv́). These specifications are contrastive even though the downstepped H is more restricted. Some contrasts between H- and L-toned monosyllabic verb roots are given in (30).

(30) H vs. L-tone contrast in Dagaare (AB:445)

a. dà
buy.PFV
'buy'

b. kyɛ̀
cut.PFV
'cut'

c. bú
measure.PFV
'measure'

d. kɔ́
farm.PFV
'farm'

The downstepped H-tone occurs underlyingly on some frequent monosyllabic words, such as the demonstrative *↓ná* in (31-a) and the verb *↓táá* 'to have' in (31-b). This is, however, a rare occurrence, as most downstepped H-toned syllables do not occur in word-initial position.

(31) Downstep in frequent monosyllabic words

a. à dɔ̀ɔ̀-wóg-kpóng-↓fáá ↓ná
DEF man-tall-big-bad.SG DEM
'that tall big bad man' (AB:449)

b. ʊ́ ↓táá=ʊ̀
3SG.HORT have=3SG
'He should have it.' (AB:452)

The stem layer includes number markers on nouns and the imperfective marker on verbs. As can be seen in (32), the inverse number suffix *-rí* surfaces as H-toned after an L-toned root as in (32-a) but as L-toned if it follows a H-toned root as in (32-b). This patterns follows if this suffix is assumed to be underlyingly H-toned but undergoes dissimilation to L after another H-tone. The imperfective suffix *-rV́* has exactly the same distribution as is shown in (33). We hence conclude that there is dissimilation at the stem layer in Dagaare.

(32) Dissimilation of the number suffix ① (AB:445)

a. wɛ̀rí
wɛ̀-rí
farm-PL
'farms'

b. wɛ́g-rì
wɛ́g-rí
log-PL
'logs'

(33) Dissimilation of the imperfective suffix ① (AB:445)

a. dàà-rá
dàà-rá
buy-IPFV
'buy'

b. búúrò
búú-ró
measure-IPFV
'measure'

At the word layer, this pattern changes. All H-toned suffixes undergo downstep if they attach to a H-toned stem. According to Anttila & Bodomo (2022), this includes the focus marker on verbs[5] as well as pronominal suffixes. In (34), downstep in OCP contexts is shown for the verbal focus marker. *-la* follows the perfective suffix in (34-a), which bears an L-tone on the surface, and therefore surfaces with a H-tone.[6] In (34-b) on the other hand, it surfaces with a downstepped H-tone because it follows a H-toned root. A different allomorph occurs on two copulas in (34-c) and (34-d) and exhibits the same pattern.

(34) Downstep for the verbal focus marker ②

a. ʊ̀ dì-èè=lá
3SG eat-PFV-FOC
'She has eaten' (AB:455)

b. ʊ̀ kúlí-ꜜlá yí-rì
3SG go.home.PFV-FOC house-SG
'He has gone home' (AB:460)

c. à dɔ́ɔ́ ì-ɛ́ kʊ́ɔ́rꜜ-áá
DET man.SG COP-FOC farmer-SG
'The man is a farmer' (Ang:13)

d. à kʊ̰̀ɔ̰́ wáá-ꜜɛ́ tʊ́lꜜʊ́ŋ
DET water COP-FOC hot
'The water is hot' (Ang:13)

In compounds, we also find a downstep anytime two H-toned roots are combined. Compounding is thus also taken to be a word layer process. In

[5] Anttila & Bodomo (2022) refer to the focus marker and the pronominal markers as enclitics but only show their effects on verbs. In fact, a similar focus marker on nouns is crucially not considered to be a clitic. I will assume that these marker are affixal in the sense that they are part of the word stratum.

[6] According to Anttila & Bodomo (2022) this form involves a more complex derivation by tonal absorption. Crucially for the present analysis, the focus marker can occur as H-toned after an L-toned suffix and follows the aspect marker.

(35) the singular adjective form *fáá* occurs with a H-tone after an L-toned root (35-a) and with a downstepped H-tone after H-toned roots (35-b).

(35) Downstep within compounds ② (AB:450+452)

a.	wὲ-fáá farm bad.SG 'bad farm'	b.	wέg-ꜜfáá log bad.SG 'bad log'
c.	kyúú-ꜜfáá moon bad.SG 'bad moon'	d.	yí-ꜜfáá house bad.SG 'bad house'

At the phrase layer, finally, adjacent H-tones are tolerated, quite similar to the pattern we saw for Shona. In (36-a) and (36-b) two H-tones are adjacent across a word boundary with neither dissimilation nor downstep applying. Note that this is not due to a ban on word-initial downstep because certain word come with a lexically specified downstep (cf. the demonstrative -ꜜ*na* in (31)).

(36) No OCP repair at the phrase layer ③ (AB:455)

a. à dɔ́ɔ́ pɔ́gɔ́
DET man woman
'the man's woman'

b. à bíí-rí fáá-rí fáá-rí
DET child-PL bad-PL bad-PL
'the children are very bad'

5.2.2 HLT account

The interstratal differences in Dagaare OCP contexts discussed in the preceding subsection are once again summarized in (37).

(37) OCP contexts in Dagaare

①	Dissimilation
②	Downstep
③	No repair

This section shows how this pattern falls out from the same basic assumption made for Shona in subsection 5.1.2: All tones decay predictably at each optimization. The crucial differences between both languages emerge since different gradient and categorical constraints are relevant. Most notably, two OCP-constraints are relevant in Dagaare, one being repaired only by dissimilation whereas the other is satisfied by both dissimilation and downstep. More concretely, gradient OCP-$\bullet$ penalizes – as in Shona – two adjacent H-toned TBU's whereas OCP-H penalizes any pair of H-tones that are adjacent on the tonal tier (38-b). Recall from (2) that we take dissimilation of H into L to be the combination of deleting an underlying H-tone and inserting an epenthetic L-tone, penalized by categorical ‖Max-H‖ (38-c) and categorical ‖Dep-L‖ (38-d) respectively. Downstep, on the other hand, is taken to be the insertion of an L-tone that remains floating between the two H-tones and hence only penalizes ‖Dep-L‖. This repair, however, only avoids a violation of OCP-H whereas OCP-$\bullet$ is still violated by this structure. How these constraint violations predict the distribution of different OCP-reactions across layers is shown in the following tableaux. As in the account for Shona, there are multiple constraints which are taken to be unviolable in Dagaare and exclude other imaginable repairs. An example would be the simple deletion of a H-tone without insertion of an L-tone, excluded by the high weight of Specify which penalizes any TBU that is not associated to a tone.

(38) a. OCP-H 10
Assign -$\frac{x+y}{2}$ violation for every pair of adjacent H-tones H_x and H_y.
b. OCP-$\bullet$ 9
Assign -$\frac{x+y}{2}$ violation for every pair of H-tones H_x and H_y that are associated to adjacent TBU's.
c. ‖Max-H‖ 6
Assign -1 violation for every input H-tone without a corresponding output tone.
d. ‖Dep-L‖ 5
Assign -1 violation for every output L-tone without a corresponding input tone.

The first tableau (39) optimizes two H-tones that are underlyingly associated to adjacent TBU's at the stem layer. Such a configuration arises, for example, if a H-toned root combines with the H-toned inverse number suffix (32). All candidates show the predictable activity decay of underlying $H_{1.0}$ to output $H_{0.8}$. Candidate (39-a) that faithfully realizes both H-toned TBU's as such hence violates both OCP-constraints by -0.8. Candidate (39-b) is the downstep repair that results from inserting an L-tone between the two H-tones that remains floating.[7] This candidate violates categorical ‖Dep-L‖ by -1 but avoids a violation of OCP-H since the two H-tones are not adjacent on the tonal tier anymore. However, this structure still violates OCP-• because two H-toned TBU's are once again adjacent. The dissimilation candidate (39-c) that deletes the second H-tone and associates an epenthetic L-tone, on the other hand, avoids both OCP-violations and trades them against a categorical ‖Dep-L‖ and ‖Max-H‖ violation. Since the weight of ‖Max-H‖ and ‖Dep-L‖ is comparably lower than the one of OCP-H and OCP-•, candidate (39-c) emerges as optimal.

(39) ① Stem: Dissimilation of H to L

$\begin{array}{cc} H_1 & H_1 \\ \vert & \vert \\ \sigma & \sigma \end{array}$	OCP-H	OCP-•	‖Max-H‖	‖Dep-L‖	$\mathcal{H}$
$\mathcal{W}$ =	10	9	7	5.5	
a. $\begin{array}{cc} H_{.8} & H_{.8} \\ \vert & \vert \\ \sigma & \sigma \end{array}$	-0.8	-0.8			-14.4
b. $\begin{array}{ccc} H_{.8} & L_1 & H_{.8} \\ \vert & & \vert \\ \sigma & & \sigma \end{array}$		-0.8		-1.0	-12.7
☞ c. $\begin{array}{cc} H_{.8} & L_1 \\ \vert & \vert \\ \sigma & \sigma \end{array}$			-1.0	-1.0	-12.0

At the word layer, the input H-tones that were already optimized at the stem layer are even weaker and enter the optimization with an input activity of 0.8 that decays to 0.6 in the output. This abstract tableau corresponds to a compound context where two stems form a new word (35).[8] Consequently, the two gradient OCP-constraints are violated to a lesser degree in candidates (40-a) and (40-b) than they were in (39-a) and (39-b). The system has now reached a 'sweet spot'. Neither the fully faithful candidate in (40-a) nor the

[7]For better readability, we also exclude the low-weighted markedness constraints that this structure violates, most notably *Float-L that penalizes any unassociated L-tones.

[8] If a word layer suffix like the verbal focus marker (34) or a pronominal suffix is added to a stem, the input in fact contains an asymmetric distribution of activity: $/H_{0.8}/ + /H_1/$. This follows since there is no independent reason to assume that word layer suffixes are optimized on their own prior to concatenation; in contrast to the assumption we made for Shona in section 5.1. For such inputs, the OCP-violations induced by candidates a. and b. increase from -0.6 to -0.7 – our constraint weights still predict the same candidate b. as optimal.

candidate (40-c), which repairs the violation of both OCP constraints by dissimilation, can become optimal. Instead, the downstep candidate (40-b) is predicted to be optimal: It still violates OCP-$\bullet$ but avoids a violation of the categorical ‖MAX-H‖ and hence fares better than (40-c).

(40) ② Word: Downstep

$\mathrm{H}_{.8}$ $\mathrm{H}_{.8}$ / σ σ $\mathcal{W}$ =	OCP-H 10	OCP-$\bullet$ 9	‖MAX-H‖ 7	‖DEP-L‖ 5.5	$\mathcal{H}$
a. $\mathrm{H}_{.6}$ $\mathrm{H}_{.6}$ / σ σ	-0.6	-0.6			-11.4
☞ b. $\mathrm{H}_{.6}$ L_1 $\mathrm{H}_{.6}$ / σ σ		-0.6		-1.0	-10.9
c. $\mathrm{H}_{.6}$ L_1 / σ σ			-1.0	-1.0	-12.5

At the phrase layer, the tones have decayed once more and the OCP violations are consequently even less severe than they were in (40).[9] This predicts that candidate (41-a) without any repair strategies becomes optimal since the categorical violations of ‖DEP-L‖ (41-b) or ‖DEP-L‖ and ‖MAX-H‖ (41-c) are too serious in comparison.

[9] If a word ending in a H-toned word layer suffix is preceding a word beginning with a H-tone, the input would again be asymmetrical: /…$\mathrm{H}_{0.8}$/ + /$\mathrm{H}_{0.6}$…/. As before (cf. footnote 8), the assumed constraint weights still make the correct prediction of an optimal candidate a. at the phrase layer, even if the OCP-violations of candidates a. and b. increase to -0.5 instead of -0.4.

(41) ③ Phrase: No OCP-repair

$\underset{\sigma}{\text{H}_{.6}}\ \ \underset{\sigma}{\text{H}_{.6}}$ $\mathcal{W}=$	OCP-H 10	OCP-• 9	‖MAX-H‖ 7	‖DEP-L‖ 5.5	$\mathcal{H}$
☞ a. $\underset{\sigma}{\text{H}_{.4}}\ \ \underset{\sigma}{\text{H}_{.4}}$	-0.4	-0.4			-7.6
b. $\underset{\sigma}{\text{H}_{.4}}\ \text{L}_1\ \underset{\sigma}{\text{H}_{.4}}$		-0.4		-1.0	-9.1
c. $\underset{\sigma}{\text{H}_{.4}}\ \ \underset{\sigma}{\text{L}_1}$			-1.0	-1.0	-12.5

The crucial weighting arguments for these three different behaviors at the stem, word, and phrase layer are summarized in (42). They compare the two thresholds between dissimilation and downstep (42-a) and the one between downstep and toleration of the OCP (42-b).

(42) Crucial weighting relations: Dagaare

a. Dissimilation at ① but downstep at ②

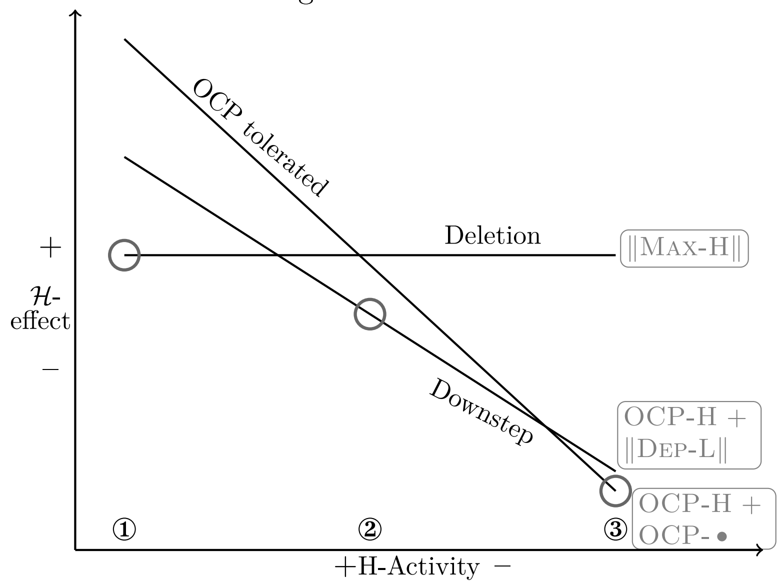
0.8×OCP-•	>	1×‖MAX-H‖			(39) c>b
		1×‖MAX-H‖	>	0.6×OCP-•	(40) b>c

b. Downstep at ② but toleration of the OCP at ③

0.6×OCP-H	>	1×‖DEP-L‖			(40) b>a
		1×‖DEP-L‖	>	0.4×OCP-H	(41) a>b

As in Shona, the main interaction is the one between categorical and gradient constraints. In Dagaare, two OCP-violations become less and less important until they can finally be tolerated at the phrase layer. On the other hand, the most unfaithful dissimilation candidate c. violates two categorical faithfulness constraints and hence keeps its harmony score constant throughout all layers. Candidates b. with a downstepped H-tone as a repair have a more intricate effect on harmony since the relevant constraints include one with a constant violation score and one with a variable violation score that depends on gradiently decaying activity. Its slope in (43) is consequently inbetween the other two candidates and it can become optimal at the word layer.

(43) HLT thresholds in Dagaare



5.3 No repair or dissimilation in Bari

Our third case study is the Eastern Nilotic language Bari. All the following data is taken from the thorough description of Bari tone in Yokwe (1986) (='Y'). The tonal patterns in Bari are highly relevant for any discussion of restricting interstratal differences since they violate the Strong Domain Hypothesis. As we will see below, OCP violations are tolerated within the stem and only repaired at the word and phrase layer via dissimilation.

5.3.1 Data

The TBU in Bari is taken to be the syllable that can bear H (=v́), L (=v̀), or a rising LH contour (=v̌). A falling HL contour is only possible on final syllables. The most striking lexical aspect about tone in Bari is that nouns and verbs show a completely different tone pattern. Whereas for nouns, the majority of imaginable tonal combinations are attested (44), verbs can only be specified as H and LHL (45).[10]

[10]Monosyllabic nouns with a /LHL/ specification surface as H-toned. That their tonal specification is /LHL/ and not /H/ becomes clear in a variety of tonal behaviours; most notably the spreading of the tonal melody to the tone-less benefactive suffix /-kIn/ (Yokwe,

(44) Nouns: Unrestricted tones (Y:122-129)

a.	kí	‘heaven’	/H/
b.	kák	‘earth’	/L/
c.	têng	‘group, herd’	/HL/
d.	wúrí	‘wild pig’	/H.H/
e.	dùpàʔ	‘cradle’	/L.L/
f.	dúlùr	‘castor oil plant’	/H.L/
g.	bòngóʔ	‘cloth’	/L.H/

(45) Verbs: Two tone melodies (Y:12+13)

a.	/H/		b.	/LHL/	
	kí	‘to cook’		ng’í	‘raise’
	kúr	‘to borrow’		dòdông	‘to shake’
	bóró	‘to smear’		dìlílì	‘to winnow grain’

We start our discussion of tonal processes at the phrase layer where dissimilation applies and changes any H-tone that follows another H-tone to L. This is shown in (46) for nouns following H-final verbs and in (47) for the infinitive construction which combines two verbs.

(46) Nouns undergo dissimilation triggered by a preceding verb ③ (Y:207)

a. /pòní à ’dép kópò/ — cf. kópò ‘cup’
pòní à ’dép kòpò ‘Poni held the cup’

b. /jàdà à nyású mángà/ — cf. mángà ‘mangoe’
jàdà à nyású màngà ‘Jada ate mangoes’

(47) Verbs undergo dissimilation triggered by a preceding verb ③ (Y:386)

a. /nân nyànyár gwójà/ — cf. gwó-j-à ‘to dance’
nân nyànyár gwòjà ‘I like to dance’

b. /nân dèdén wúrjà/ — cf. wúr-j-à ‘to write’
nân dèdén wùrjà ‘I know how to write’

Another context for dissimilation at the phrase layer can be found in the associative constructions that links two noun phrases together and establishes a relationship between those two NP’s; a prominent one being a possession relation. In Bari, the second NP2 in an associative construction surfaces with one of three prefixes: /ló-/ MASC, /ná-/ FEM.SG, /tí-/ PL which agree with the head noun of NP1[11]. A simple example for an associative construction

1986, 18-24).

[11] Those are described as ‘particles’ in Yokwe (1986) but taken to be (phrasal) prefixes

without any tonal changes is given in (48).

(48) Associative construction: No tonal changes
/ng'úrò ló-jàdà/
ng'úrò ló-jàdà 'the child of Jada' Y:249

In (49), however, the associative prefix is preceded by a H-final NP1 and surfaces as L-toned: It hence underwent dissimilation to avoid two adjacent H-tones at the phrase layer.

(49) NP1 triggers dissimilation on an associative prefix ③

a.	/kéré ló-pòní/		
	kéré lò-pòní	'the gourd of Poni'	Y:252
b.	/kòng'é ló jàdà/		
	kòng'é lò jàdà	'the eye of Jada'	Y:258
c.	/àmúlèrèjín tí wàní/		
	àmúlèrèjín tì-wàní	'the flutes of Wani'	Y:259
d.	/rét ná bòngó?/		
	rét nà bòngó?	'the tear of the cloth'	Y:259

The associative prefixes are hence undergoers of dissimilation at the phrase layer. Within the word, the associative prefixes can also be triggers of dissimilation and cause the H-tone of a following NP2 to dissimilate to L. Associative prefixes are hence analysed as word layer affixes here. This can be seen in the examples (50) where NP1 ends in an L-tone. In contrast to (49), the associative prefix can hence faithfully realize its underlying H-tone. A H-initial NP2, however undergoes dissimilation. In (50-a+b), NP2 has an underlying HL-specification and only the initial TBU is changed; in (50-c+d), both TBU's of NP2 are H-toned and become L-toned after the associative prefix.

here.

(50) Associative prefixes trigger dissimilation on following NP2 ②

a. /kú'bâ ló-kúlàng'/
   kú'bâ ló-kùlàng' 'the in-law of Kulang' Y:254

b. /mòkêt ná-kópò/
   mòkêt ná-kòpò 'the handle of the cup' Y:254

c. /mókòt ló-wúrí/
   mókòt ló-wùrì 'the leg of the pig' Y:253

d. /píòng' tí-kídí/
   píòng' tí-kìdì 'the water of the well' Y:249

The application of dissimilation at the word and the phrase layer predicts counterbleeding configurations where a H-tone trigger of the word layer is dissimilated into an L-tone at the phrase layer. This is exactly what we find in the associative construction if a H-final NP1 is conjoined with a H-initial NP2 as in (51). At the word layer, the associative prefix triggers dissimilation for NP1 but undergoes dissimilation itself at the phrase layer, making the previous application of the process surface-opaque. This is shown in derivational terms for (51-a) in (52).

(51) Dissimilation at ③ counterbleeds dissimilation at ②

a. /kùmé ló-wúrí/
   kùmé lò-wùrì 'the nose of the pig', Y:256

b. /kònyén tí-dúlùr/
   kònyén tì-dùlùr 'the seeds of castor', Y:256

(52) Counterbleeding in (51-a)

a. Dissimilation at ② H + [H.H] –> {H.L.L}

b. Dissimilation at ③ {L.H} + {H.L.L} –> {L.H} {L.L.L}

Affixes other than the associative prefix, however, do not undergo or trigger dissimilation. This is shown in (53) with the singulative suffix that surfaces with a stable H-tone, irrespective of whether a H-final root precedes or not. In (53-a+b), it follows an L-final root and no dissimilation is expected to begin with. In (53-c+d), however, it follows an underlying /H.H/ and /L.H/ root respectively and we would expect a surface form */-tàt/ if dissimilation were operative. This state of affairs follows from the assumption that the associative prefix is a word layer affix but that singulative /-tÁt/ (and plural /-Á/ and /-Át/) are stem layer suffixes. Dissimilation is operative at the word layer but does not apply to adjacent H-tones created at the stem layer.

(53) No dissimilation at ①: Stable H-toned /tAt/ (Y:172+173)

	ROOT	SG	
a.	yàng'ò	yàng'ò-tát	'yaws'
	gìlà	gìlà-tát	'white man'
b.	rímà	rímà-tát	'blood'
	rúbì	rúbì-tát	'molar tooth'
c.	bóyí	bóyí-tát	'wild fig tree'
	kúrú	kúrú-tát	'worm'
d.	kàmú	kàmú-tát	'guest'
	gùrú	gùrú-tát	'lizard'

5.3.2 HLT account

The interstratal differences for OCP-triggered H-dissimilation observed in Bari are once again summarized in (54).

(54) OCP contexts in Bari

①	No repair
②	Dissimilation
③	Dissimilation

As in Dagaare and Shona, these facts are predicted from the cyclic optimization with a single grammar and a predictable activity decay for all tones (12). The most central difference to both Shona and Dagaare is that the relevant OCP-constraint triggering the repair is a categorical one (55-a).[12] This is already key to the late repair pattern that Bari shows: OCP-violations have the same effect on the harmony score throughout layers and repairing them is only possible after tones have decayed a bit. Shona and Dagaare show the opposite pattern: OCP-violations became less and less severe until they were finally tolerated in the last layer. Consequently, at least one of the faithfulness constraints penalizing dissimilation, MAX-H (55-b), is gradient in Bari. As before, the following tableaux only consider candidates that become optimal in OCP contexts at some layer: Toleration of the OCP and dissimilation

[12] Other facts about Bari tonology make clear that the relevant OCP constraint cannot be formulated about H-toned TBU's. More concretely, the language also employs a process of binary H-spreading that can can result in the de-association of an underlying L-tone. Crucially, there is no OCP-repair in this context – a fact which receives an independent explanation if one considers the de-associated L-tone to remain floating between the two otherwise adjacent H-tones.

in Bari. All other candidates are taken to be excluded by higher-weighted constraints. An example for an obvious competitor would be deletion of the second instead of the first H-tone: As before, we take this option to be excluded by some directional constraint.

(55) a. ‖OCP-H‖
Assign -1 violation for every pair of adjacent H-tones not associated to the same TBU.

b. MAX-H
Assign -x violation for every input H_x that is not realized in the ouput.

c. ‖DEP-L‖
Assign -1 violation for every output L-tone without a corresponding input tone.

Tableau (56) starts with a stem layer optimization. The candidate realizing both underlying H-tones (with their predictable activity decay of 0.2) (56-a) becomes optimal since a full -1 violation of ‖OCP-H‖ is better than the combined violations of MAX-H and ‖DEP-L‖ that the dissimilation candidate (56-b) induces.

(56) ① Stem: No OCP-repair

H_1 H_1 / σ σ	‖OCP-H‖ 12	MAX-H 10	DEP-L 3	$\mathcal{H}$
☞ a. $H_{.8}$ $H_{.8}$ / σ σ	-1			-12
b. $H_{.8}$ L_1 / σ σ		-1	-1	-13

At the word layer (57), however, the gradient violations of MAX-H induced if an input $H_{0.8}$ is deleted are already less fatal than the categorical ‖OCP-H‖ violation and dissimilation (57-b) hence becomes optimal. It is important to point out that the activity levels of the input tones at the word layer are asymmetric: The initial H-tone necessarily belongs to the associative prefix – the only H-toned element that can be added at the word layer – which enters the derivation with its full lexical activity (cf. footnote 8 that a similar context is possible in Dagaare). The grammar nevertheless correctly predicts dissimilation simply because it always affects the second of two H-tones.

(57) ② Word: Dissimilation of H to L

$\underset{\sigma}{\overset{H_1}{\mid}}$ $\underset{\sigma}{\overset{H_{.8}}{\mid}}$	‖OCP-H‖ 12	MAX-H 10	DEP-L 3	$\mathcal{H}$
a. $\underset{\sigma}{\overset{H_{.8}}{\mid}}$ $\underset{\sigma}{\overset{H_{.6}}{\mid}}$	-1			-12
☞ b. $\underset{\sigma}{\overset{H_{.8}}{\mid}}$ $\underset{\sigma}{\overset{L_1}{\mid}}$		-0.8	-1	-11

Tableau (58) adds a phrase layer optimization with an input of two H-tones with the decayed activity of 0.6. Since we already established in (57) that a -0.8 violation of MAX-H is already better than a categorical ‖OCP-H‖ violation, it should come as no surprise that this even weaker tone will also delete to avoid an OCP violation in optimal (58-b). Similarly to the asymmetric input in (57), a phrase layer input could also be /... $H_{0.6}$/ + /$H_{0.8}$.../ if a H-final word precedes a word starting with an associative prefix (cf. (50)). In such a case, the optimization would mirror the one given for the word layer where a -0.8 violation of MAX-H is tolerated in order to avoid a -1 violation of ‖OCP-H‖.

(58) ③ Phrase: Dissimilation of H to L

$\underset{\sigma}{\overset{H_{.6}}{\mid}}$ $\underset{\sigma}{\overset{H_{.6}}{\mid}}$	‖OCP-H‖ 12	MAX-H 10	DEP-L 3	$\mathcal{H}$
a. $\underset{\sigma}{\overset{H_{.4}}{\mid}}$ $\underset{\sigma}{\overset{H_{.4}}{\mid}}$	-1			-12
☞ b. $\underset{\sigma}{\overset{H_{.4}}{\mid}}$ $\underset{\sigma}{\overset{L_1}{\mid}}$		-0.6	-1	-9

Since there are only two possible behaviours in OCP contexts in Bari, there are only two relevant weighting relations showing the threshold between toleration of the OCP or dissimilation (59).

(59) Crucial weighting relations: Bari

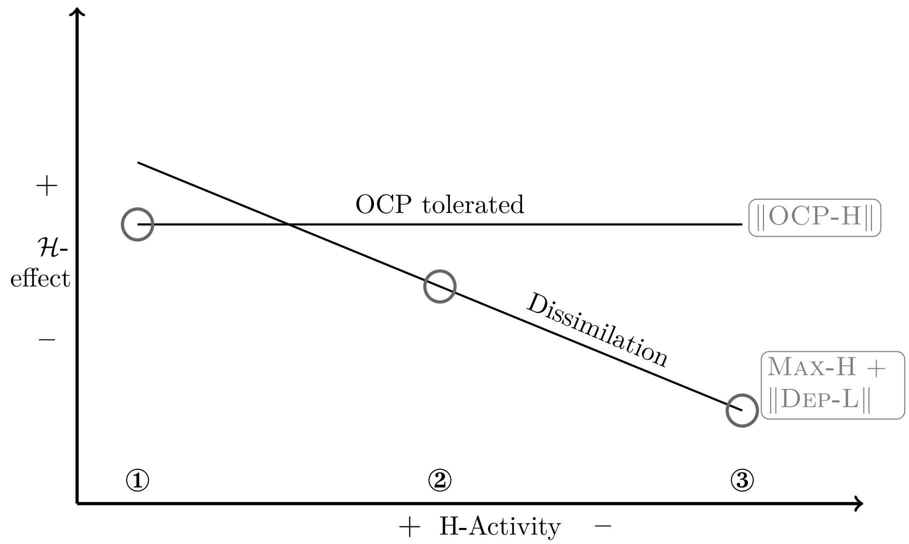
Toleration of the OCP at ① but dissimilation at ②

$1\times\textsc{Max}\text{-H} + 1\times\|\textsc{Dep}\text{-L}\| \;>\; 1\times\|\text{OCP-H}\|$ (56) a>b

$1\times\|\text{OCP-H}\| \;>\; 0.8\times\textsc{Max}\text{-H} + 1\times\|\textsc{Dep}\text{-L}\|$ (57) b>a

The graphical illustration of these thresholds is also relatively simple since it only involves one straight line for the categorical OCP-violation and one falling line for the gradient faithfulness violations induced by the dissimilation candidate. Both lines cross before the word layer where the behaviour changes.

(60) HLT thresholds in Bari



5.4 HLT as restrictive theory: No ABA patterns

In the preceding three subsections, we saw three detailed HLT accounts for three of the patterns that are part of our typology of interstratal OCP-differences. To shed further light on the predictive power of HLT, let us briefly return to the fully typology, repeated in (61).

(61) The typology of OCP-repairs

	① Stem	② Word	③ Phrase
Moro	Deletion	No repair	No repair
Shona	Fusion	Deletion	No repair
Chitonga	Deletion	Deletion	No repair
Dagaare	Dissimilation	Downstep	No repair
Tiriki	Dissimilation	Dissimilation	No repair
Moore	Dissimilation	Dissimilation	No repair
Dagbani	Dissimilation	Dissimilation	No repair
Jita	Deletion	Deletion	No repair
Chizigula	Deletion	Deletion	No repair
Kishambaa	Fusion	Downstep	Downstep
Tswana	Fusion	Fusion	Downstep
Venda	Fusion	Fusion	Dissimilation
Bari	No repair	Dissimilation	Dissimilation

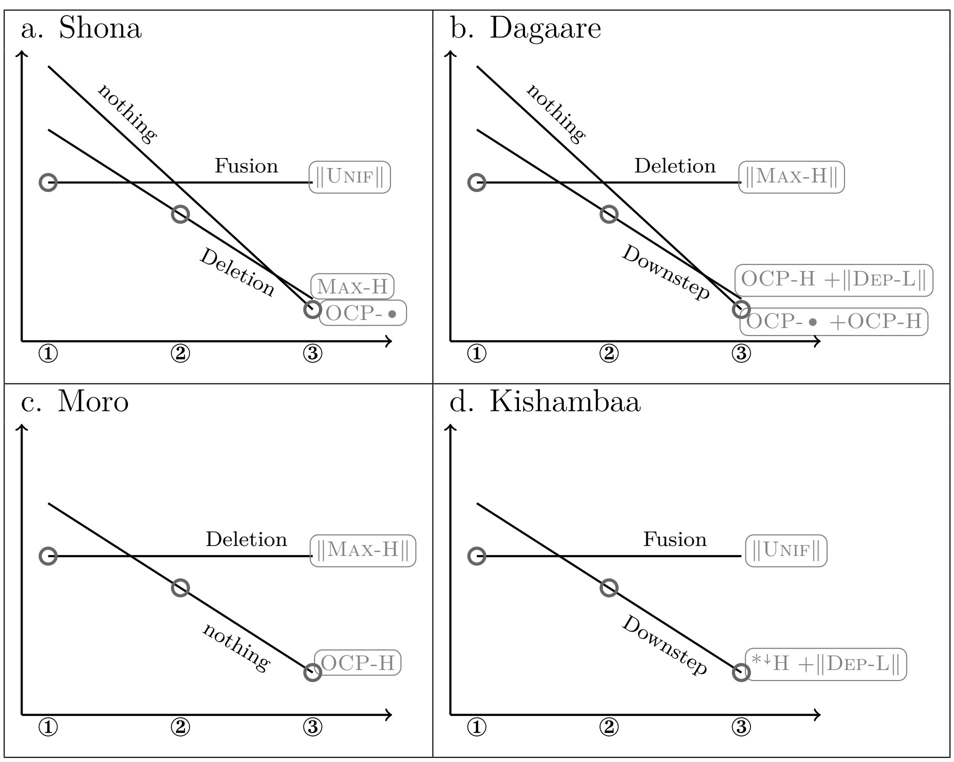
As was already discussed above in subsection 3, these 8 different patterns can be summarized into three general ones if one categorizes them according to the distribution of different phonological behaviours across layers; repeated in (62). There are either three different behaviours at ①, ②, and ③ (62-A), a certain behaviour at ① and a different one at ② and ③ (62-B), or an identical behaviour at ① and ② and a different one at ③ (62-C).

(62) General patterns of interstratal differences in our sample

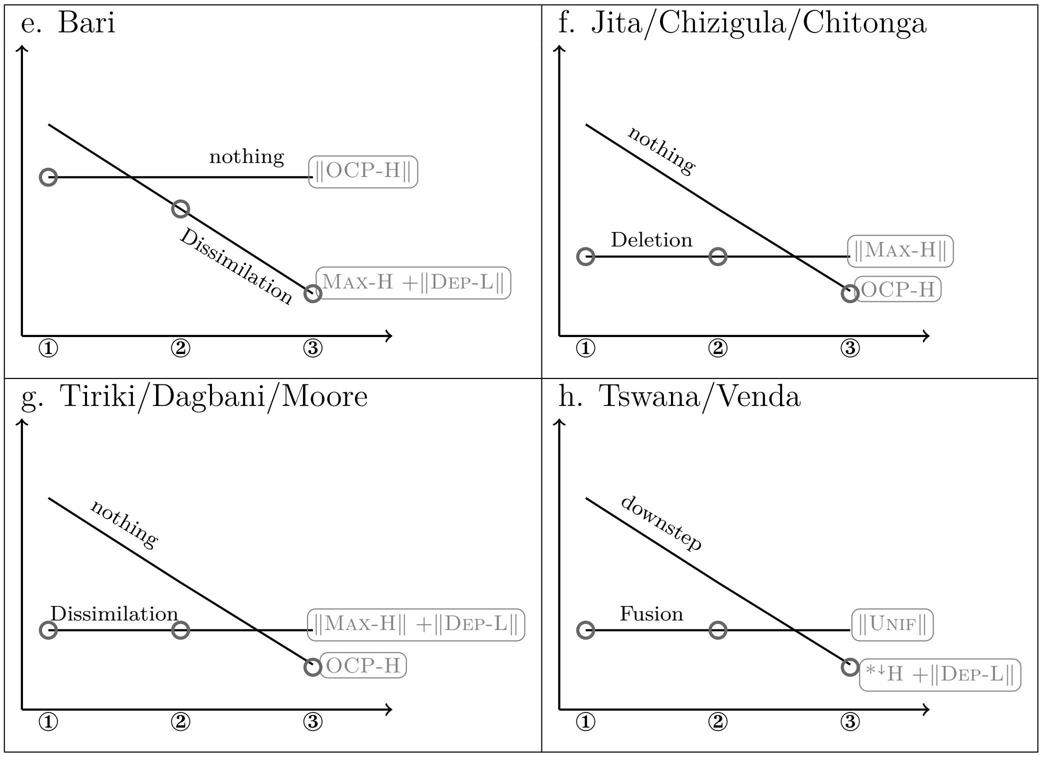
	Pattern A	Pattern B	Pattern C
①	Behaviour 1	Behaviour 1	Behaviour 1
②	Behaviour 2	Behaviour 2	Behaviour 1
③	Behaviour 3	Behaviour 2	Behaviour 2
	3 lgs	7 lgs	3 lgs

This distinction is helpful when thinking about how to analyse these patterns in HLT since every change in behaviour between layers corresponds to a crossing of harmony scores in HLT. The graphs in (63) summarize how all these patterns fall out in HLT accounts, repeating the ones we saw above for Shona, Dagaare, and Bari, and adding all the remaining languages from our sample. Crucially, the general patterns (62) result from only three general patterns of harmony score crossings in (63): Thresholds (63-a+b) for pattern (62-A), thresholds (63-c+e) for pattern (62-B), and thresholds (63-f-h) for pattern (62-C). The patterns in (63) only crucially differ in the specific constraints that are relevant in either their categorical or their gradient version. An interesting comparison is the one between the Bari (63-e) and the Tiriki/Dagbani/Moore (63-g) pattern that both show an interstratal difference between dissimilation and toleration of the OCP. However, whereas dissimilation only starts applying at layer ② in (63-e), it ceases to apply at layer ③ in (63-g): A contrast that falls out in HLT since in pattern (63-e), a categorical ‖OCP-H‖ and a gradient MAX-H is relevant whereas it is the other way around in (63-g) where a categorical ‖MAX-H‖ and a gradient OCP-H are relevant. Another contrast we want to point out is the one between Moro (63-c) and Jita/Chizigula (63-f): Both patterns show deletion at ① and toleration of the OCP at ③ but differ in the behaviour of ②. Both patterns fall out from a categorical ‖MAX-H‖ penalizing deletion and a gradient OCP-H penalizing adjacent H-tones: They simply differ in the respective weighting difference between them and hence the point where both harmony score effects cross.

(63) HLT thresholds: Our typology of OCP-repairs

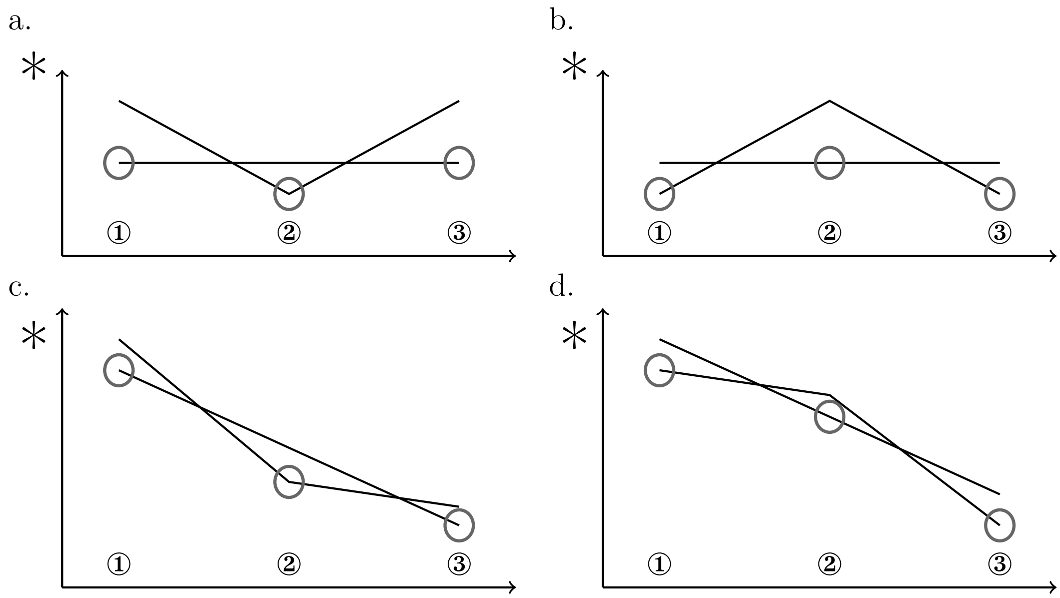


(63) HLT thresholds: Our typology of OCP-repairs, contd.



HLT is thus perfectly capable of predicting all instances of interstratal differences in OCP contexts that we found in our representative typological study. However, far more interesting are the patterns that a HLT model is inherently unable to predict. All imaginable interstratal differences that are in violation of the *ABA-restriction (6) are impossible to predict in the HLT model as we presented it here. This follows from the fact that all interstratal differences in HLT fall out from predictable activity adjustment across layers and the fact that activity adjustment is necessarily monotonic: If a grammar predicts activity adjustment (cf. (12)), then this activity adjustment is necessarily identical across layers, simply because only a single grammar optimizes repeatedly in HLT. This means that the abstract three crossing patterns that were exemplified in (63) already illustrate the realm of possible interstratal differences that HLT is able to predict: Harmony score effects are either identical across layers or fall consistently. Crucially, however, harmony score effects cannot change their direction (64-a+b), or the steepness of their fall (64-c+d).[13] This inherently excludes ABA-patterns that would require harmony score interactions as the impossible ones in (64).

(64) No ABA-patterns in HLT



6 Conclusion and discussion

This paper contributed both an empirical and a theoretical claim to the ongoing discussion of how morphology and phonology interact. Empirically, we argued for the *ABA-restriction which excludes certain imaginable interstratal differences in the languages of the world. We emphasized that this restriction is in spirit very similar to both the Stratum Domain Hypothesis and the Strong Domain Hypothesis but more restrictive than the former and less restrictive than the latter. Our argument was both based on a survey of the existing arguments for or against these alternative restrictions and a representative typological study of OCP differences across layers.

[13] Steepness stays constant for single constraints or sets of constraints. It can only differ between constraints for two reasons. First, some constraints ignore gradient activation and have a constant harmony score effect, whereas others are sensitive to gradient activation. Second, gradient markedness constraints refer to output activation while certain faitfhulness constraints, e.g. gradient MAX constraints need to refer to input activation. Both differences can potentially cause crossing patterns.

We argued that the existence of the *ABA-restriction is a major argument for HLT, a novel theory of the morpho-phonology interface where the same grammar optimizes cyclically within each language but phonological elements may decay with each optimization. HLT hence utilizes the concept of gradient activities that has been argued to predict a variety of phenomena like doubly lexically conditioned phonologies (e.g. Smolensky & Goldrick, 2016; Rosen, 2016, 2019a,b; Zimmermann, to appear), lexical exceptionality (e.g. Kushnir, 2017; Zimmermann, 2019), typological markedness asymmetries (e.g. Walker, 2019), or generalizations about reduplication patterns (Zimmermann, 2021) and explores one further prediction that this assumption makes: Combined with the assumption of cyclic optimizations, it naturally predicts interstratal differences from one and the same grammar. Furthermore, HLT inherently predicts the *ABA-restriction and is hence far more restrictive and empirically more adequate than any alternative model based on co-grammars.

We want to end with three outlooks for future research. Firstly, we argued that OCP contexts are an ideal testing ground for investigating interstratal differences across languages since there is such a wide variety of attested possible reactions to this marked configuration. There are a number of other empirical areas which are also promising for further establishing the typology of interstratal differences. We already saw an interstratal difference for H-tone spreading in Shona: H-tones spread to the next two following TBU's at the stem layer but only to the following TBU at the word and phrase layer, cf. (17). Bari also employs an interstratal difference for H-spreading: Binary H-spreading does not apply at the stem layer but it does apply at the word and the phrase layer. Our initial impression is thus that interstratal spreading asymmetries are nearly as common as OCP asymmetries, especially among Bantu languages. Outside of tones, vowel hiatus avoidance is an area which shows a similar variety in possible repairs as tonal OCP problems (e.g. Casali, 2005) and several cases have been reported to show interstratal differences in hiatus resolutions. A particularly interesting one can be found in Gua (Obiri-Yeboah & Rasin, 2024). Vowel hiatus resolution might hence be another promising empirical area where the validity of the *ABA-restriction and the argument for HLT can be further investigated.

Secondly, the HLT accounts discussed here are all based on three layers, namely stem, word, and phrase. This assumption is in line with the classical Lexical Phonology architecture and was empirically adequate for all the patterns we encountered in our typological study. Neither HLT nor its empirical prediction of the *ABA-restriction, however, are restricted to three morpho-syntactic layers. There might hence indeed be languages with interstratal differences distributed over four or even more morpho-syntactic domains that each require an optimization cycle in HLT.

A final point which should be investigated more thoroughly in the future is the validity of the Strong Domain Hypothesis or a possible HLT-equivalent. We already emphasized above that Bari (65-A) is the only pattern in our typology that violates the Strong Domain Hypothesis. Looking at all patterns in our sample again, it is in fact apparent that the vast majority of patterns (=9 out of 13) are not only in conformity with the Strong Domain Hypothesis but in fact exemplify its poster-child-pattern: Processes cease to apply at the phrase layer (65-A-C). And Bari could in fact be re-analysed as a (65-D) pattern if we assume that there is surface-identical fusion of adjacent H-tones at the stem layer instead of no repair. In contrast to Shona, there is no argument from the interaction with other processes that distinguishes the two analytical options: Both the assumption of no repair and the assumption of fusion at the stem level can be incorporated into an account that predicts the correct Bari pattern. A re-analysis based on fusion would put Bari within the predictions of the Strong Domain Hypothesis and is reminiscent of the early classical autosegmental tonal analyses where tone fusion was taken to be an automatic convention rather than a process on its own (e.g. Leben, 1973; McCarthy, 1986; Myers, 1986). Future research needs to establish whether the prominent traces of the Strong Domain Hypothesis in our typology are a coincidence or are robust tendencies about the typological patterns that need some theoretical explanation. In the latter case, the *ABA-restriction might in fact be too weak and the HLT architecture might need to be reconsidered to be even more restrictive than it already is.

(65) General patterns of interstratal differences in our sample

	A	B	C	D	E	F
①	process 1	process 1	process 1	process 1	process 1	no repair
②	no repair	process 2	process 1	process 2	process 1	process 1
③	no repair	no repair	no repair	process 2	process 2	process 1
	Moro	Shona Chizigula Dagaare	Tiriki Moore Dagbani Jita Chizigula	Kishambaa	Tswana Venda	Bari

7 Appendix: Our data

In the following paragraphs, we summarize the core generalizations with some examples for all the languages given in our typology. An exception are Shona, Dagaare, and Bari which were already discussed extensively above.

Moro (Jenks & Rose, 2011, 2015)

Jenks & Rose (2011, 216-223) show that no Moro noun can include more than one H-tone span which is taken as evidence for a constraint from the OCP family.

(66) OCP restricts possible tone patterns on nouns ① (Jenks & Rose, 2011, 216-223)

a. ómóná 'leopard'
b. *ómoná
c. áveja 'spring'
d. *ávejá

Further evidence for a high-ranked OCP comes from H-tone spreading patterns that are blocked if they would lead to an OCP violation. This can be seen with the prefix *ék-* in (67).

(67) OCP restricts spreading from locative prefix *ék-* at ① Jenks & Rose (2011, 218)

	NOUN	LOCATIVE	GLOSS
a.	ogovélá	ék-ógovél	'monkey'
b.	áppa	ég-áppa	'father'
c.	áveja	ék-áveja	'spring'

In verbs, this leads to a phonological alternation, where certain morphemes inside a sub-word domain are only realized with a H-tone if no other H-tone occurs within that domain. This sub-word domain corresponds to a (Macro-)stem and includes object agreement prefixes, the progressive prefix, durative/iterative reduplication, verb roots, and the aspect/mood suffix. An otherwise H-toned root shows up without a H-tone if preceded by the iterative/durative reduplicant in (68-a,b), or by a H-toned object prefix in (68-c,d), or if it is followed by the underlyingly H-toned perfective suffix in (68-e,f). This applies iteratively between affixes, as shown for iterative/durative reduplication plus object prefixes in (68-g), and iterative/durative reduplication and a perfective suffix in (68-h).

(68) OCP triggers tone deletion in verbs at ① (Jenks & Rose, 2011, 238.239)

a. ka-[ðə̱́w-á]
3SG.S-poke-IPFV
'he's about to poke'

b. ka-[ðá̱ð∼ðə̱w-a]
3SG.S-DUR∼poke-IPFV
'he's poking'

c. ka-[və̱́léð-a]
3SG.S-pull-IPFV
's/he's going to pull it'

d. ka-[ɲə̱́-və̱leð-a]
3SG.S-1SG.O-pull-IPFV
's/he's going to pull me'

e. k-a-[ŋá̱l-á]
3SG.S-FIN-yawn-IPFV
's/he is about to yawn'

f. k-a-[ŋa̱l-ó̱]
3SG.S-FIN-yawn-PFV
'he had yawned'

g. ka-[ŋó-ða̱ð-ðə̱w-a]
3SG.S-3SG.O-DUR∼poke-IPFV
'he's poking her'

h. ka-[ða̱ð-ðə̱w-ó]
3SG.S-DUR∼poke-PFV
'he was poking her'

The repair is not found with other affixes in the verbal domain, such as subject agreement prefixes, tense prefixes, and object agreement suffixes. This is shown for a combination of past *Cá*-reduplication and a root H-tone in (69-a), for a subject marker and a root H-tone in (69-b), and for a subject marker combined with past *Cá*-reduplication and a root H-tone in (69-c).

(69) No OCP repair at ② (Jenks & Rose, 2011, 239,241)

a. ŋá̲∼ŋ-[á̲ff-a]
PST∼3SG.NC.S-[shoot-IPFV]
'He was going to shoot.'

b. é̲-g-[á̲ff-a]
PST∼3S.NC-shoot-IPFV
'I am about to shoot.'

c. é̲-gá̲-g-[á̲ff-a]
1SG.S-PST∼-NC-shoot-IPFV
'I was going to shoot.'

Jenks & Rose (2015, 295) also provide examples of H-tones that become adjacent at the phrase layer across a word boundary. These are not repaired. The OCP is thus not active on the phrase layer anymore.[14]

(70) No OCP repair at ③ (Jenks & Rose, 2015, 295)

a. /á-g-a-toð-ó n-a-ɲé-aləŋ-ət̪-e/→[ágatoðó nʌ́ɲʌləŋət̪i]
2SG.S-CL-R-get.up-P C-2SG.S-1SG.O-sing-AP-C.P
'you got up and sang to me'

b. /g-a-toð-ó n-óŋó-ɲé-aləŋ-ət̪-e/ → [katoðó nóŋɲ↓ʌ́ləŋət̪i]
CL.S-R-get.up-P C-3SG.S-1SG.O-sing-AP-C.PFV
's/he got up and sang to me'

Chitonga (Bickmore & Mkochi, 2018)

Chitonga has a complex array of OCP resolution strategies depending on the syllabic, moraic, and morphosyntactic structure of a form as described in detail in Bickmore & Mkochi (2018). In the unmarked case, where two monomoraic H-toned vowels become adjacent, the repairs differ between deletion at the stem/word-level and no deletion in phrase-level contexts. Inside the stem, the second of the two tones is deleted. This can be seen in the example in (71), where a H-toned root is combined with a final vowel suffix that bears a so-called melodic H-tone.

[14] AP=applicative, C=complementizer, CL=noun class, C.P=consecutive perfective, O=object marker, P=perfective, R=root clause, S=subject marker.

(71) Deletion at ① (Bickmore & Mkochi, 2018, 7)

a. kù-[nám-á] → kù-nàám-à
INF-lie-FV
'to lie'

b. kù-[nám-á] cáà → kù-nàm-á cáà
INF-lie-FV NEG
'to not lie'

Across a stem boundary, a slightly different deletion pattern is found. Instead of the second tone, the first tone is deleted. This change is obscured by a rule of tone doubling that applies in non-final position. If no deletion would apply, then at least the syllable after the root would be expected to be linked to a H-tone. This can be seen with the H-tone of the itive prefix *cí-* in (72).

(72) Deletion across a stem boundary at ② (Bickmore & Mkochi, 2018, 14)

a. ti-ngu-cí-[púm-is-an-a] cáà →
1PL.S-PST-ITI-beat-CAUS-REC-FV NEG
ti-ngu-cí-púm-is-an-a cáà

'We did not go and cause each other to beat.'

b. ti-ngu-cí-[nám-is-an-á] cáà→ ti-ngu-cí-nám-is-an-á
1PL.S-PST-ITI-lie-CAUS-REC-FV NEG
'We did not go and cause each other to lie.'

Across a word boundary, no such deletion applies, as shown in (73). This is shown with H-toned noun roots adjacent to a H-toned negation particle in (73).

(73) No deletion across words at ③ (Bickmore & Mkochi, 2018, 6)

a. mù-tú cáà
NC-head NEG
'not a head'

b. mà-só cáà
NC.PL-eye NEG
'not eyes'

In other morphophonological contexts, OCP violations are also repaired by tone retraction or downstep. In certain contexts, violations are tolerated even inside words.

Tiriki (Paster & Kim, 2011)

In Tiriki (Bantu, Kenya), a H-tone dissimilates to an L-tone after a H-tone inside stems and words (Paster & Kim, 2011). This domain includes the verb root and any kind of prefix or proclitic, i.e. any word internal domain. Examples with object agreement prefixes and H-toned root are given in (74), between object prefixes and tense prefixes in (75).

(74) Dissimilation between a prefix and a verb root at ① (Paster & Kim, 2011, 79)

a. xu-mú-rhúmul-il-a → xúmúrhùmulila
INF-3SG.O-hit-APPL-FV
'to hit for him'

b. xu-mú-rhúm-a → xúmúrhùma
INF-3SG.O-send-FV
'to send him/her'

c. y-a-gá-rhúm-a → y-á-gá-rhùm-a
3SG.S-GENER.PST-send-FV
'he sent it (cl. 5; general past)'

(75) Dissimilation between object prefixes and tense prefixes ② (Paster & Kim, 2011, 79)

a. v-áá-rhḿul-il-an-a → váárhùmulilana
3PL.S-EXP-hit-APPL-RECIP-FV
'they have hit for each other before (experiential)'

b. y-áá-xárag-a zi-nguza → yááxàraga zinguza
3PL.S-EXP-cut-FV NC-vegetable
'he has cut vegetables before'

(76) Dissimilation between proclitics and prefixes at ② (Paster & Kim, 2011, 79)

a. ní=zí-nguvú → nízìngúvù
with=NC-hippo
'with hippos'

b. ní=lí-duumá → nílìdúúmà
with=NC-corn
'with corn (sg.)'

Dissimilation also applies between prefixes and proclitics, which we take to be word-level prefixes, as shown in (76). Dissimilation, however, does not apply between words. There is an orthogonal phonologically derived environment effect in that H-tones that become adjacent through spreading undergo downstep. Therefore, the tolerated sequence of two H-tones is highlighted in the sentence in (77).

(77) No dissimilation between words at ③ (Paster & Kim, 2011, 86)

a. à-zì-hééz-ꜜáá zí-ꜜsíím**bwá lí**-ꜜdúúmà
3SG-3SG.O-give-FV NC-dog NC-corn
'he is giving dogs corn'

b. à-heez-áá zí-ꜜsíímbwá zí-ꜜmwáá**mú lí**-ꜜdúúmà
3SG-give-FV NC-dog NC-black NC-corn
'he is giving black dogs corn'

Moore (Kenstowicz et al., 1988; Kenstowicz, 1994)

Kenstowicz et al. (1988) argue that the tonal surface patterns arise from assuming underlying H, L, and tone-less TBU's, a dissimilation process and H-spreading. In (78), the underlyingly H-toned number suffix *-gó* and *dó* retain their H after an L-final root (78-a), dissimilate to an L after a H-final root (78-b), and spread their H leftwards to provide underlyingly tone-less roots with a tone (78-c).

(78) Dissimilation at ①/② (Kenstowicz et al., 1988, 78)

	SG	PL	
a.	kor-gó	kor-dó	'sack'
	roo-gó	ro-tó	'house'
b.	wób-go	wób-do	'elephant'
	láŋ-go	lán-do	'hole'
c.	bíd-gó	bí-tó	'sorrel'
	míu-gú	míi-dú	'red'

There is no dissimilation at the phrase layer anymore. There are not too many contexts where this can clearly be seen since the suffixes are lost in non-phrase final position: There are hence many downsteps arising from the deletion of L-toned suffixes. One context are noun+adj constructions where the noun suffix is realized on the adjective and not the noun (79-a); another is the imperative+noun construction (79-b). In both cases, a sequence of two

H-tones is tolerated.

(79) No dissimilation at ③ (Kenstowicz et al., 1988, 80,83)

a. sá bɛ́d-a
broom big-PL
'big brooms' cf. (kor bɛ́d-a 'big sacks')

b. zá sáa-ga
bring broom-SG
'bring a broom' (cf. kõ sáa-ga 'give a broom')

Dagbani (Hyman, 1993)

In Dagbani, inside words two adjacent H-tones result in a sequence of a low and a H-tone. This contrasts with a spread-out H-tone. This can be seen in vocative forms, such as in (80) even though this is obfuscated by the application of a postlexical left-to-right H-tone spreading process. In most contexts, this dissimilation is rendered opaque by the deletion of the resulting L-tone as contour tones only occur in a restricted set of contexts.

(80) Dissimilation in Dagbani at ①/② (Hyman, 1993, 240)

a. pág-aa → pág-áá
woman-VOC
'Woman!'

b. sánH-aa → sán-áà
stranger-VOC
'Woman!'

Across word boundaries, on the other hand, adjacent H-tones are tolerated. As spreading also applies at the phrase level, its application also does not trigger dissimilation. The application of H-tone spreading obscures the effect of word-internal dissimilation in phrase-final position in (81-b).

(81) No dissimilation at ③ (Hyman, 1993, 237,238)

a. sán tɨ́ta-lí → sáːn tɨ́tálí
stranger big-NC
'big stranger'

b. pag-á yí-lí → pág-á yí-lí
woman-NC house-NC
'woman's house'

Jita (Downing, 1996, 2014)

Downing (2014) shows that a H-tone is deleted when adjacent to another H-tone. This is true for all word-internal morpheme combinations even though the directionality depends on the specific affix combination. The surface structure is obscured by phrase-layer tone shift, which shifts all H-tones a single syllable to the right, except for the penultimate and final syllable of a phrase. Therefore, the single H-tone resulting from dissimilation and originating from the object prefixes in (82), shows up on the underlyingly H-toned root. With word-level prefixes, such as the negative prefix *tá-*, a similar pattern is observed in (83).

(82) Deletion of H-tones on roots after object prefixes at ① (Downing, 2014, 104)

a. oku-mu̲-βó̲na
INF-3SG.O-see
'to see him/her'

b. oku-mu̲-βó̲nera
INF-3SG.O-get.for
'to get for him/her'

c. oku-βá̲-síí̲ndika
INF-PL.O-push
'to push you pl./them'

(83) Deletion of H-tones after negative prefix at ② (Downing, 2014, 104,105)

a. a-tá-a-βóna → ataaβóna
3SG.S-NEG-PST.TODAY-see
's/he did not see'

b. a-tá-a-saanga → ataasáánga
3SG.S-NEG-PST.TODAY-meet
's/he did not meet'

c. a-tá-a-βá-saanga → ataaβásaanga
3SG.S-NEG-PST.TODAY-PL.O-meet
's/he did not meet them (Cl.2)'

(84) Deletion of H-tone on hesternal marker before H-toned roots at ② (Downing, 2014, 105)

a. /a-amá-gósorera/ → aamagosórera
3SG.S-HEST.PST-visit
's/he visited (someone)'

b. /a-amá-kú-gósorera/ → aamakugósorera
3SG.S-HEST.PST-2SG.O-visit
's/he visited you (sg.)'

(85) Deletion of H-tones adjacent to distant future marker at ② (Downing, 2014, 105)

a. /a-tá-lí-jí-βá-léétera/ → atalijíβaleetera
3SG.S-NEG-DST.FUT-NC10.O-NC2.O-bring
's/he will not bring them (Cl. 10) to them (Cl. 2)'

b. /a-tá-lí-jí-mú-gusisya/ → atalijímugusisya
3SG.S-NEG-DST.FUT-NC10.O-NC1.O-bring
's/he will not sell them (Cl. 10) to him/her (Cl.1)'

The polysyllabic morphemes in (84), show the effects of tone shift more clearly, since the root H-tone shows up on the second root syllable after dissimilating the hesternal past prefix *amá-* H-tone in (84-a) but the object prefix H-tone instead shows up on the first root syllable in (84) after causing dissimilation of both the root and the hesternal past prefix H-tone. In (85), the distant future marker *lí-* shifts its H-tone on the following object marker and no other underlying H-tone surfaces.

Tone shift is crucially conditioned by phrasal context and can result in adjacent H-tones. It only applies to H-tones that are not in a the final or penultimate syllable of the phonological phrase. (86-b) shows that OCP violations are tolerated at the phrase layer even if they result from tone shift. (86-a) shows that H-tones can occur in this phrasal position.

(86) Phrase-medial tone shift at ③ (Downing, 2014, 103)

a. oku-βoná iiɲoɲi
INF-see bird
'to see a bird'
(cf. oku-βóna 'to see'; iiɲoɲi 'bird')

b. oku-lya múnó
INF-eat a.lot
'to eat a lot'
(cf. oku-lyá 'to eat'; munó 'a lot')

Chizigula (Kenstowicz & Kisseberth, 1990)

Chizigula stems, i.e. a domain consisting of suffixes, roots, and object prefixes, undergo a specific kind of dissimilation. When two H-tones would occur in this domain, only one surfaces in the expected position. This position is not the underlying position of either H-tones but instead the penultimate syllable.

(87) Dissimilation at ① (Kenstowicz & Kisseberth, 1990, 172)

a. na-[wá-lúlunganya] → na-wa-lulungánya
1SG.S-3.PL.O-take.advantage
'I am taking advantage of them'

b. na-[wá-hángalasanyiza] → na-wa-hangalasanyíza
1SG.S-3.PL.O-carry.many.things.for
'I carry many things for them'

In a slightly larger domain, which also includes the subject prefixes, a downstep occurs if two H-tones are forced to be adjacent. This can be seen with the third person subject prefix in (88).

(88) Downstep between a subject prefix and a root at ② (Kenstowicz & Kisseberth, 1990, 170)

INF	ku-lombéza	ku-tabana	ku-fundika
3SG.S	a-[lóm↓béza]	a-[tá↓bána]	a-[fún↓díka]
gloss	'ask'	'weave a spell'	'tie a knot'

At the phrase level, the same downstep is found if a word ends in a H-tone and the following word starts with a H-tones, as in (89).

(89) Downstep at ③ (Kenstowicz & Kisseberth, 1990, 181,188)

a. n-a-on-á ↓kúli
1SG.S-1SG.O-see-FV dog
'I see a dog'

b. n-a-fis-á ↓tági
1SG.S-3SG.O-hide-FV egg
'I am hiding an egg'

c. n-a-kom-á ↓kúwi
1SG.S-1SG.O-kill-FV turtle
'I am killing a turtle

d. nambikila mndélé ↓nkhánde
1SG.S.cook girl food
'I am cooking the girl food'

Kishambaa (Odden, 1982)

In Kishamba, downstep applies between words, as well as inside word if two H-tones occur adjacent to each other either underlyingly or due to spreading. This is shown in (90) for the word-internal domains and in (91) for a context across word boundaries.

(90) Downstep at ② (+HSpread)
[ní-kí]-[↓chí-kóm-á]
1SG-PROG-it-kill-FV
'I was killing it' (O2:191)

(91) Downstep at ③
ní ↓kúi
COP dog
'it is a dog' (O2:187)

No such downstep is found between the object marker and the root (92), which according to Odden, "form a tighter phonological and morphological unit", i.e. a stem. The H-tone on roots and on object prefixes undergo phonological processes together, e.g. Downstep in (90).

(92) Fusion between Obj+root at ①
[ku]-[wá-kóm-á]
INF-them-kill-FV
'to kill them' (O2:191)

Tswana (Zerbian, 2015; Zerbian & Kügler, 2015, 2023)

Zerbian & Kügler (2015) present a phonetic study showing that Tswana only shows downstep across word boundaries within phonological phrases. This same pattern is also described for Southern Sotho (Khoali). In Tswana, no downstep is found between H-tones inside a word, such as between a H-toned verbal prefix and a H-toned roots in (93).

(93) No downstep inside words at ①/② (Zerbian & Kügler, 2023, 379)

a. Ba-rwá bá-thúsá ba-lemi.
NP2-son SC2-help NP2-farmer
'The sons help the farmers.'

b. Bɔ́-mmɛ́ bá-bɔ́ná ba-lé.
NP2A-mother SC2-see NP2-girl
'The mothers see the uninitiated girls.'

Across word boundaries on the other hand, downstep is found, as shown in (94). Since it affects whole spans of H-tones that derive either from spreading or from previously adjacent H-tones, the assumption has been that adjacent H-tones fuse at the stem-level and at the word-level.

(94) Downstep across word boundaries at ③ (Zerbian & Kügler, 2023, 374,378,379)

a. (Ke bítsá ꜜpódi bosígo)$_\phi$
1SG call NP9.goat evening
'I call the goat in the evening

b. Ba-ná ꜜbá-tshábá ma-kgóa.
NP2-child SC2-fear NP6-white.person
'The children fear the white people.'

c. Ba-ná ꜜbá-a-bítsa.
NP2-child SC2-ASP-call
'The children are calling.'

Venda (Cassimjee, 1992)

In Venda, Meeussen's Rule, i.e. dissimilation, applies to nouns that occur following a H-tone at the phrase level. After an L-tone or in isolation, some nouns show up with a H-tone on the root, whereas following a H-final word they show up with a falling tone. Cassimjee (1992) analyzes this as an instance of dissimilation obscured by H-tone spreading. The presence of a falling tone, i.e. an L-tone component, indicates that this is dissimilation and not deletion of H-tones. An example is shown in (95).

(95) Dissimilation in Venda nouns at ③ (Cassimjee, 1992, 17)

a. mu-sélwa → mù-sélwà
NC-bride
'bride'

b. ndi-vhóná mu-sélwa → ndi-vhóná mú-sêlwà
1SG-see NC-bride
'I see the bride.'

Cassimjee (1992) assumes that inside stems, fusion takes place between H-tones. The principal reason for this is that these H-tones undergo phrase-level dissimilation as if they were one H-tone, as shown for root-internal H-tones in the example in (96). If dissimilation applied to two distinct H-tones, we might expect only the first one to undergo it in (96-b). A general application of dissimilation inside words is excluded by the isolation form in (96-a), which shows to adjacent H-toned syllables.

(96) Stem-level fusion feeds Dissimilation at ① (Cassimjee, 1992, 17)

a. mu-sádzí → mu-sádzí
NC-bride
'woman'

b. ndi-vhóná mu-sádzí → ndi-vhóná mu-sâdzì
1SG-see NC-woman
'I see the woman.'

This argument is strengthened because similar tonal configurations are also found across prefix boundaries. The potential prefix *ngá* in (97-a) and the past tense prefix *ó-* in (97-b) and the negative prefix *thí* bear a H-tone and can precede a H-toned root or prefix without triggering any repair process.[15]

(97) No dissimilation inside words at ② (Cassimjee, 1992, 120,136,174)

[15](Cassimjee, 1992, 174,175) actually assumes that dissimilation can apply inside words but will rendered opaque by specific contour simplication rule that only applies to pre-penultimate syllables and emulates the effects of fusion by mapping a sequence of two H-tones to a doubly linked H-tone. If dissimilation were to apply between prefixes, this would mean that in our typology it would count as exhibiting an ABB pattern, where fusion only applies at the stem level.

a. ndi-ngá-[vhóná]
1SG-POT-see
'I may see.'

b. nd-ó-[vhóná]
1SG-PST-see
'I saw.'

c. a-thí-ngá-[gidími]
1SG-NEG-POT-run
'I may not run.'

References

Angsongna, Alexander. 2023. *Aspects of the morphophonology of dagaare*: University of British Columbia dissertation.

Anttila, Arto & Adams Bodomo. 2022. Tone and morphological level ordering in Dagaare. *Phonology* 39(3). 443–471.

Bermúdez-Otero, Ricardo. 2012. The architecture of grammar and the division of labour in exponence. In Jochen Trommer (ed.), *The morphology and phonology of exponence*, 8–83. Oxford: Oxford University Press.

Bickmore, Lee & Winfred Mkochi. 2018. OCP effects in Malawian CiTonga tone patterns. *Nordic Journal of African Studies* 27.

Bobaljik, Jonathan. 2012. *Universals in comparative morphology. suppletion, superlatives, and the structure of words.* Cambridge, Mass.: MIT Press.

Caha, Pavel. 2009. *The nanosyntax of case*: Universitetet i Tromsø dissertation.

Casali, Roderic. 2005. *Resolving hiatus.* Ann Arbor, MI: UMI.

Cassimjee, Farida. 1992. *An autosegmental analysis of Venda tonology.* New York, London: Garland.

Downing, Laura. 1996. *The tonal phonology of Jita.* München: Lincom.

Downing, Laura. 2014. Melodic verb tone patterns in Jita. *Africana Linguistica* 20. 101–119.

Faust, Noam & Paul Smolensky. 2017. Activity as an alternative to autosegmental association. talk given at mfm 25, 27th May, 2017.

Graf, Thomas. 2019. Monotonicity as an effective theory of morphosyntactic variation. *Journal of Language Modelling* 7(2). 3–47.

Graf, Thomas. 2020. Monotonicity in syntax. In Dun Deng, Fenrong Liu, Mingming Liu & Dag Westerståhl (eds.), *Monotonicity in logic and language*, 35–53. Berlin, Heidelberg: Springer.

Hammarström, Harald, Robert Forkel & Martin Haspelmath. 2019. Glottolog 4.0. https://glottolog.org/ accessed 2019-11-18, Max Planck Institute for the Science of Human History, Jena.

Hyman, Larry. 1993. Structure preservation and postlexical tonology in Dagbani. In Sharon Hargus & Ellen M. Kaisse (eds.), *Studies in lexical phonology*, 235–254. San Diego: Academic Press.

Inkelas, Sharon & Cemil Orhan Orgun. 1995. Level ordering and economy in the lexical phonology of Turkish. *Language* 71. 763–793.

Inkelas, Sharon & Cheryl Zoll. 2007. Is grammar dependence real? A comparison between cophonological and indexed constraint approaches to morphologically conditioned phonology. *Linguistics* 45(1). 133–171.

Jang, Hayeun. 2019. Emergent phonological gradience from articulatory synergies: simulations of coronal palatalization. talk, presented at the LSA 2019, New York, January 05, 2019.

Jenks, Peter & Sharon Rose. 2011. High tone in Moro: effects of prosodic categories and morphological domains. *Natural Language and Linguistic Theory* 29. 211–250.

Jenks, Peter & Sharon Rose. 2015. Mobile object markers in Moro: The role of tone. *Language* 269–307.

Jones, Patrick. 2014. *Tonal interaction in Kinande: Cyclicity, opacity, and morphosyntactic structure*. Cambridge, MA: Massachusetts Institute of Technology dissertation.

Kenstowicz, Michael. 1994. *Phonology in generative grammar*. Cambridge MA: Blackwell.

Kenstowicz, Michael & Charles Kisseberth. 1990. Chizigula tonology: The word and beyond. In and Sharon Inkelas & Draga Zec (eds.), *The phonology-syntax connection*, 163–194. Chicago and London: University of Chicago Press.

Kenstowicz, Michael, Emmanuel Nikiema & Meterwa Ourso. 1988. Tonal polarity in two Gur languages. *Studies in the Linguistic Sciences* 18. 77–104.

Kiparsky, Paul. 1982. From cyclic phonology to lexical phonology. In Harry van der Hulst & Norval Smith (eds.), *The structure of phonological representations*, 131–175. Dordrecht: Foris.

Kiparsky, Paul. 1984. On the lexical phonology of Icelandic. In Claes-Christian Elert, Iréne Johansson & Eva Strangert (eds.), *Nordic prosody ii: Papers from a symposium*, University of Umea.

Kiparsky, Paul. 1985. Some consequences of lexical phonology. *Phonology yearbook* 2(1). 85–138.

Kiparsky, Paul. 2015. Stratal OT: A synopsis and FAQs. In Yuchau E. Hsiao & Lian-Hee Wee (eds.), *Capturing phonological shades within and across languages*, 1–45. Newcastle: Cambridge Scholars Publishing.

Kushnir, Yuriy. 2017. Accent strength in Lithuanian. talk, given at the workshop on Strength in Grammar, Leipzig, November 12, 2017.

Leben, William. 1973. *Suprasegmental phonology*: MIT dissertation.

Legendre, Géraldine, Yoshiro Miyata & Paul Smolensky. 1991. Harmonic grammar: A formal multi-level connectionist theory of linguistic well-formedness: An application. In *Proceedings of the twelfth annual conference of the cognitive science society*, 884–891. Lawrence Erlbaum.

Legendre, Géraldine, Antonella Sorace & Paul Smolensky. 2006. The optimality theory-harmonic grammar connection. In Paul Smolensky & Géraldine Legendre (eds.), *The harmonic mind: From neural computation to optimality-theoretic grammar*, 339–402. Cambridge, MA: MIT Press.

McCarthy, John. 1986. OCP effects: Gemination and antigemination. *Linguistic Inquiry* 17. 207–63.

McCarthy, John J. 2010. An introduction to harmonic serialism. *Language and Linguistics Compass* 4(10). 1001–1018.

McCarthy, John J. 2016. The theory and practice in harmonic serialism. In John J McCarthy & Joe Pater (eds.), *Harmonic grammar and harmonic serialism*, 47–87. Sheffield, UK: Equinox.

Mohanan, Karuvannur Puthanveettil. 1982. *Lexical phonology*: MIT dissertation.

Mohanan, Karuvannur Puthanveettil. 1986. *The theory of lexical phonology*. Reidel Publishing Co., Dordrecht, Holland.

Myers, Scott. 1986. *Tone and the Structure of Words in Shona*: University of Massachusetts, Amherst dissertation.

Myers, Scott. 1997. OCP effects in Optimality Theory. *Natural Language and Linguistic Theory* 15(4). 847–892.

Obiri-Yeboah, Michael & Ezer Rasin. 2024. Productive phrasal opacity in Gua: A challenge to Stratal Optimality Theory. *Natural Language and Linguistic Theory* DOI https://doi.org/10.1007/s11049–024–09615–7.

Odden, David. 1982. Tonal phenomena in Kishambaa. *Studies in African Linguistics* 13. 177–208.

Paster, Mary & Yuni Kim. 2011. Downstep in Tiriki. *Linguistic Discovery* 9. 71–104.

Pater, Joe. 2009. Weighted constraints in generative linguistics. *Cognitive Science* 33. 999–1035.

Potts, Christopher, Joe Pater, Karen Jesney, Rajesh Bhatt & Michael Becker. 2010. Harmonic grammar with linear programming: From linear systems to linguistic typology. *Phonology* 77–117.

Rosen, Eric. 2016. Predicting the unpredictable: Capturing the apparent semi-regularity of rendaku voicing in Japanese through Harmonic Grammar. In Emily Clem, Virginia Dawson, Alice Shen, Amalia Horan Skilton, Geoff Bacon, Andrew Cheng & Erik Hans Maier (eds.), *Proceedings of BLS 42*, 235–249. Berkeley: Berkeley Linguistic Society.

Rosen, Eric. 2019a. Evidence for gradient input features from Sino-Japanese compound accent. In *Proceedings of amp 2018*, LSA.

Rosen, Eric. 2019b. Learning complex inflectional paradigms through blended gradient inputs. Proceedings of the 2019 meeting of the Society for Computation in Linguistics, January 2019, NYU.

Smith, Peter W, Beata Moskal, Ting Xu, Jungmin Kang & Jonathan David Bobaljik. 2019. Case and number suppletion in pronouns. *Natural Language & Linguistic Theory* 37. 1029–1101.

Smolensky, Paul & Matthew Goldrick. 2016. Gradient symbolic representations in grammar: The case of French liaison. Ms, Johns Hopkins University and Northwestern University, ROA 1286.

Tebay, Sören E. 2023. Kinande tone as layered and gradient. talk, given at the mfm30, University of Manchester, 26.05.2023, URL: https://home.uni-leipzig.de/tebay/tebay2023Kinande.pdf.

Tebay, Sören E. 2025. The lexical-postlexical watershed, *ABA, and the weak domain hypothesis. talk, given at the DGfS workshop Variation in Cyclicity, 06.03.2025, online available at https://home.uni-leipzig.de/tebay/Tebay2025ABA.pdf.

Trommer, Jochen. 2019. A Harmonic Layer account of Levantine Arabic syncope. talk, presented at OCP 16, Verona, January 17, 2019; online available at https://home.uni-leipzig.de/jtrommer/papers/LevantineOCPVerona2019.pdf.

Trommer, Jochen. 2025. Strata. Ms. Leipzig University.

Trommer, Jochen. t.a. Strata. In *Cambridge handbook of phonology*, Cambridge University Press.

Walker, Rachel. 2019. Gradient feature activation and the special status of coronals. talks, presented at PΦF 2019, April 05, 2019.

Yip, Moira. 2002. *Tone*. Cambridge University Press.

Yokwe, Eluzai M. 1986. *The tonal grammar of Bari*: University of Illinois at Urbana-Champaign dissertation.

Zerbian, Sabine. 2015. Adjacent high tones at word- and phrase level in Tswana. talk at the workshop 'Morphosyntactic Triggers of Tone: New Data and Theories', Leipzig University.

Zerbian, Sabine & Frank Kügler. 2015. Downstep in Tswana (Southern Bantu). In *International congress of phonetic sciences*, .

Zerbian, Sabine & Frank Kügler. 2023. Sequences of high tones across word boundaries in Tswana. *Journal of the International Phonetic Association* 53.

Zimmermann, Eva. 2018. Being exceptional is being weak: Tonal exceptions in San Miguel el Grande Mixtec. In Gillian Gallagher, Maria Gouskova & Sora Heng Yin (eds.), *Proceedings of AMP 2017*, http://dx.doi.org/10.3765/amp: LSA.

Zimmermann, Eva. 2019. Gradient symbolic representations and the typology of ghost segments. In Katherine Hout, Anna Mai, Adam McCollum, Sharon Rose & Matthew Zaslansky (eds.), *Proceedings of AMP 2018*, https://doi.org/10.3765/amp: LSA.

Zimmermann, Eva. 2021. Faded copies: Reduplication as distribution of activity. *Glossa: A Journal of General Linguistics* 6. 58.

Zimmermann, Eva. 2024. The fate of tone-less TBU's in Zacatepec Chatino across morphological layers. ms., University of Leipzig, lingbuzz/008917.

Zimmermann, Eva. 2025. Interstratal differences and the *ABA-restriction: An argument for Harmonic Layer Theory. invited talk at the OCP 22, Amsterdam, February 5, 2025, URL: http://www.evazimmermann.org/uploads/1/1/7/2/117248671/ocp_slides2.pdf.

Zimmermann, Eva. to appear. Cooperation at the morpho-phonology interface: An argument for phonological locality. *Proceedings of WCCFL 2023* .

Zimmermann, Eva & Jochen Trommer. 2022. Getting weaker across layers: The tonal phonology of Shona without stratal re-ranking. invited (online) talk at the University of Warsaw, linguistics colloquium, June 17th, 2020, URL: http://www.evazimmermann.org/uploads/1/1/7/2/117248671/cyclictoneshonacorr.pdf.